



Ultima7

Workforce Data Collection Terminal



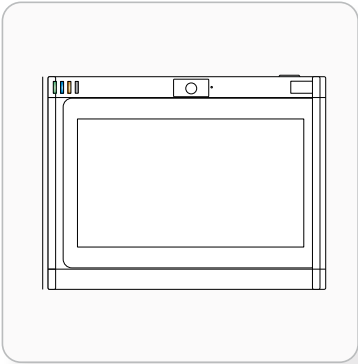
Installation Guide

This guide provides installation instructions only. For information regarding actual operation and configuration of the **Ultima7** Time & Attendance terminal, refer to the user manual.

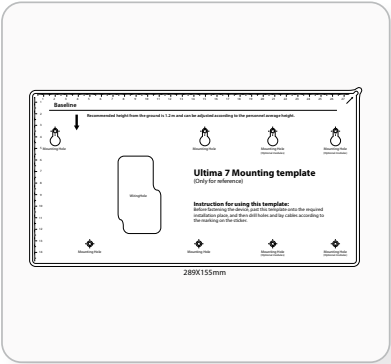


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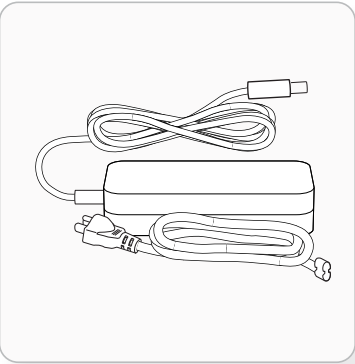
Packing List



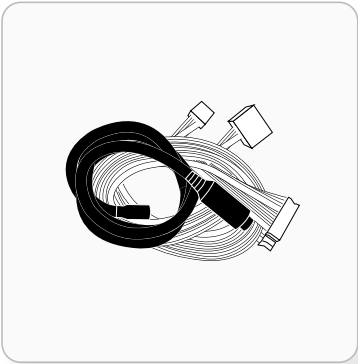
Base



Mounting Template



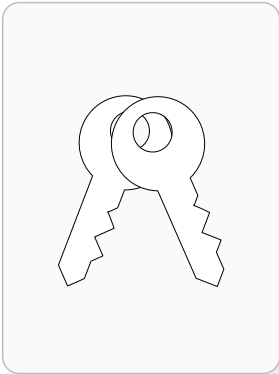
Power Supply



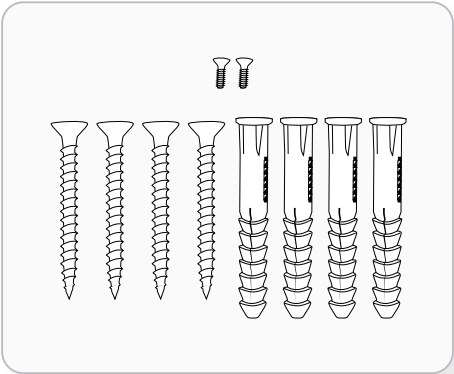
Power Cable



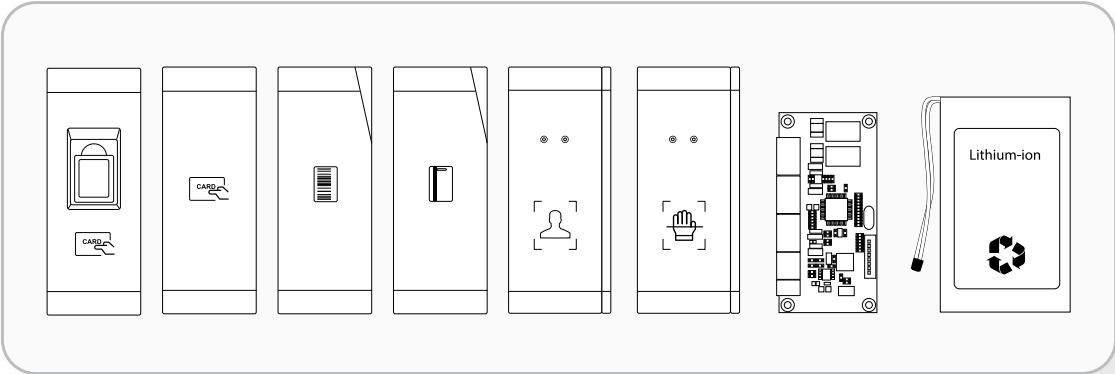
Network Cable



Keys(2)



Screws & Anchors



Optional Modules

The contents of the package may vary depending on the modules that you choose.

Pre-Installation

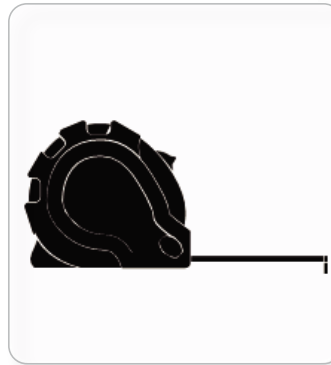


Important

To avoid damaging the main board, always wear an antistatic wrist strap when touching the main board in any way, including connecting and disconnecting wires.

- Do not remove the cover assembly from its anti-static packaging until you are ready to install the device
- Save the paper mounting template and any other printed documents

Note: Because of the small size of certain screws, we recommend that you use small screwdrivers with magnetic tips.



Measuring Tape



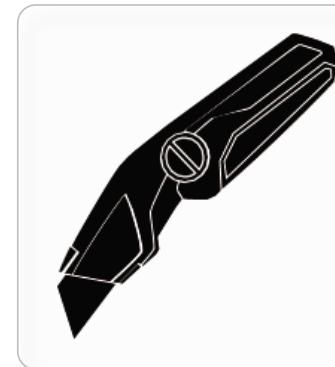
Level



Wire Cutter



Screwdrivers



Drywall Knife

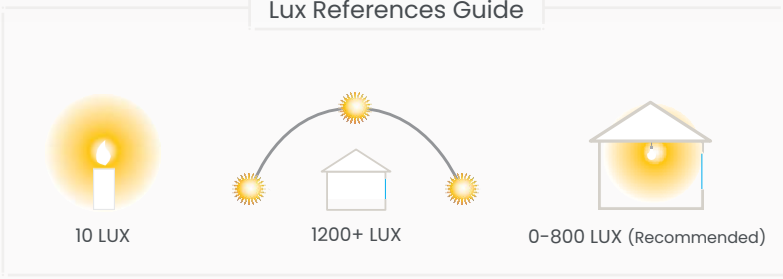
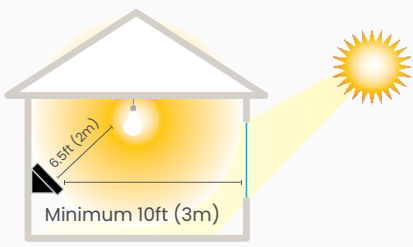


Power Drill



Location

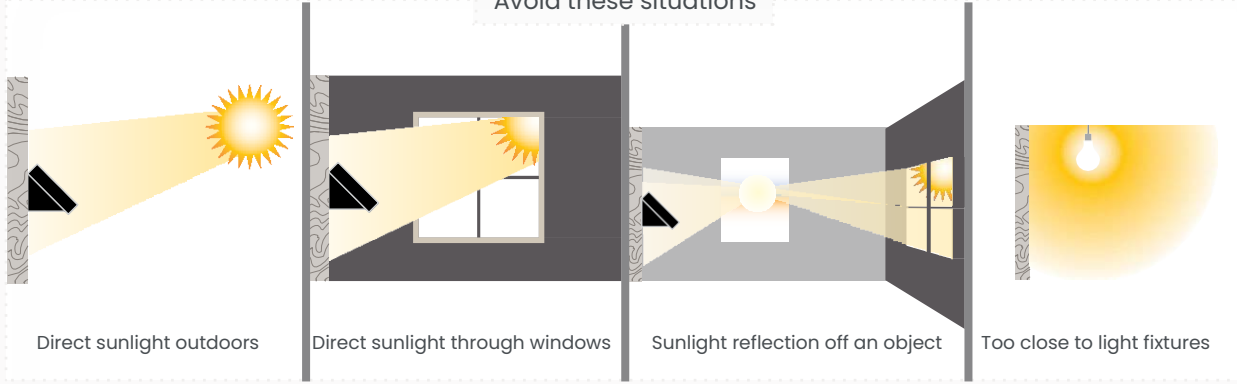
- Ultima 7 is designed for any internal environment. In rare cases, user sites may contain agents in nature that can corrode the electronics assemblies in these devices. In the event of this situation, users must move the devices to a suitable environment
- Install the Ultima 7 in an area where the screen is not exposed to direct sunlight or other high-intensity lighting that could make the screen difficult to read



Lux References Guide

Recommended Locations

The device should only be installed indoors. The illuminance of the external light source is 0-800LUX. Direct sunlight should be avoided. Install the device at least 10 feet (3m) away from windows, and 6.5 feet (2m) away from light fixtures.



Avoid these situations

Direct sunlight outdoors Direct sunlight through windows Sunlight reflection off an object Too close to light fixtures

Avoid direct sunlight, indirect sunlight reflected off mirrors, or bright artificial light.



Warning Relay is rated for **12V only** and **should not exceed the recommended voltage**. If connecting a higher voltage device some extra components maybe necessary. Please consult ZKTeco WFM for further guidance before connecting to higher voltages.



Required Personnel

Depending on the modules that you purchased, and the implementation planned for your site, you will require the following:

- A licensed electrician to install or move AC outlets and power cables
- A qualified person to run Ethernet cable

One or more qualified installers to assemble the device and mount it on the wall (a second person may be necessary to hold the cover while the other person attaches components to the cover and base).



Mounting Template

The packaging box includes a paper mounting template that you use to mark the wall with the following measurements and indicators:

- Where to screw the base of the unit into the wall
- Where cables enter the unit through access holes in the wall



Power Sources and Connections

Requirements:

- The AC power source must be grounded 100 to 240V AC, 50/60Hz input voltage
- A licensed electrician must ensure that low-voltage wiring is routed securely away from high-voltage wiring

Power source options:

The following power supply methods are available:

- Mounted near an AC outlet, with power supply plugged in
- Power-over-Ethernet+ (PoE+)
 - Warning** Power up your device with AC adapter or Power-over-Ethernet+ (PoE+). Do not connect simultaneously (It may damage the device)
- Backup battery



Ethernet Connection

- Location - Plan to install the Ultima 7 in a location that allows an easy and secure connection to an external network (for example, using the Ethernet cable)
- All cables must be in place before installing the Ultima 7

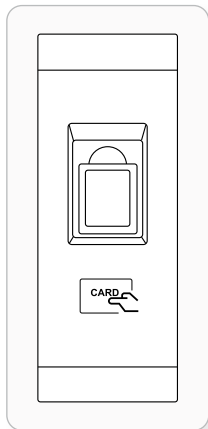


If you plan to install certain optional modules to the Ultima 7, follow the step-by-step instructions in this guide. Note the following:

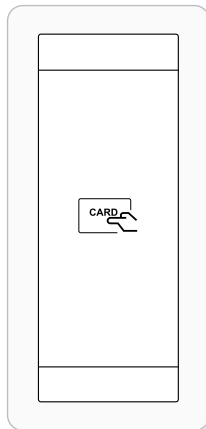
- The optional modules have components and cables that must be attached to the Ultima 7 in a certain order during specific stages of the installation
- You can only choose one between the barcode and magnetic modules

The modules combination can be as follows:

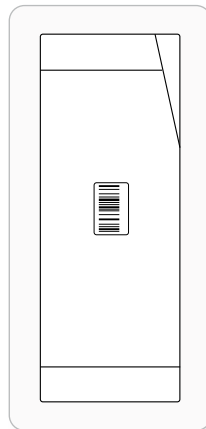
- Base+ Fingerprint & RFID module
- Base+ RFID module
- Base+ Facial Recognition module
- Base+ Palm Recognition module
- Base+ Barcode/Magnetic module
- Base+ Fingerprint & RFID module+ Barcode/Magnetic module
- Base+ RFID module+ Barcode/Magnetic module
- Internal Relay Module
- Internal PoE+ Module



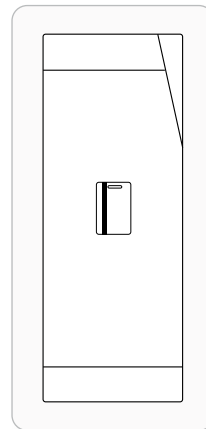
Fingerprint &
RFID module



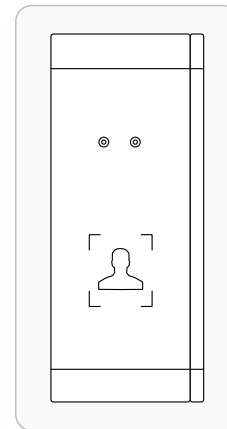
RFID
module



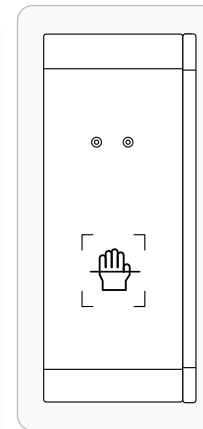
Barcode
module



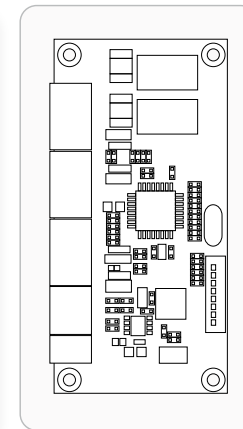
Magnetic
module



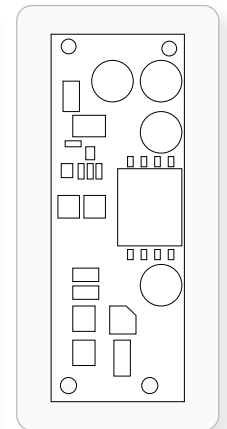
Facial
module



Palm
module



Relay
module



PoE+
module



Battery



Determine the power source for your site

Before you start the installation, plan with your electrician (if required for the power option that you choose) how the Ultima 7 will connect to the power at your site.

Choose from the following options and review the associated notes.

Power options

- Install near an AC outlet
Select a location where the distance from the wiring hole of the Ultima 7 is not more than 5 feet (1.5 meters) from the AC outlet
Note: Installing the device near an AC outlet does not protect against the AC power cord from being deliberately or inadvertently unplugged from the outlet.
- Power-over-Ethernet+ (PoE+)
The device has a PoE+ module in it. You can supply power to the device by connecting a network cable to the PoE+ switch
- Backup battery
When the power is off or in emergency, the backup battery can supply power to the device

Notes :

- A licensed electrician must install the AC outlet and associated wiring before you install the Ultima 7
- Do not connect the Ultima 7 power line to circuits that are being used for electrical devices that draw large amounts of power, such as air conditioning units, electrical motors, and compressors
- Avoid running a communications cable near devices that interfere with data transmission

Installation



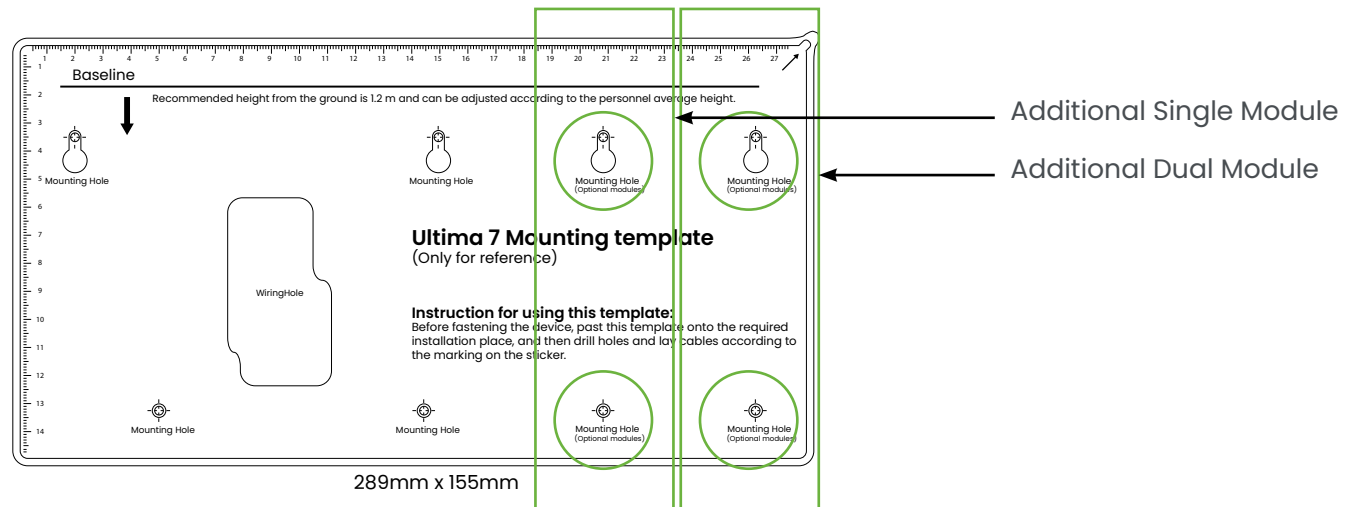
Mark insertion points for the base

The mounting template is printed on a sheet of paper that is included in the package. Use that template to mark the areas of the wall where you plan to install the Ultima 7.

Mark the wall

1. Tape the mounting template at the optimal height and location for the Ultima 7
 - a. The baseline is recommended 1.2m above the floor and can be adjusted to the actual situation
 - b. Use a level to ensure that the top of the template is horizontal to the floor
2. Mark the location of the mounting screw holes
3. If you plan to run an Ethernet cable through the wall to enter the Ultima 7 from the back or through the top, use the template to mark the area of entry
4. If you are installing the Ultima 7 near an AC outlet, be sure the outlet is within 5 feet (1.5 meters) of the unit
5. After you finish marking all screw holes and wiring hole, remove the mounting template from the wall

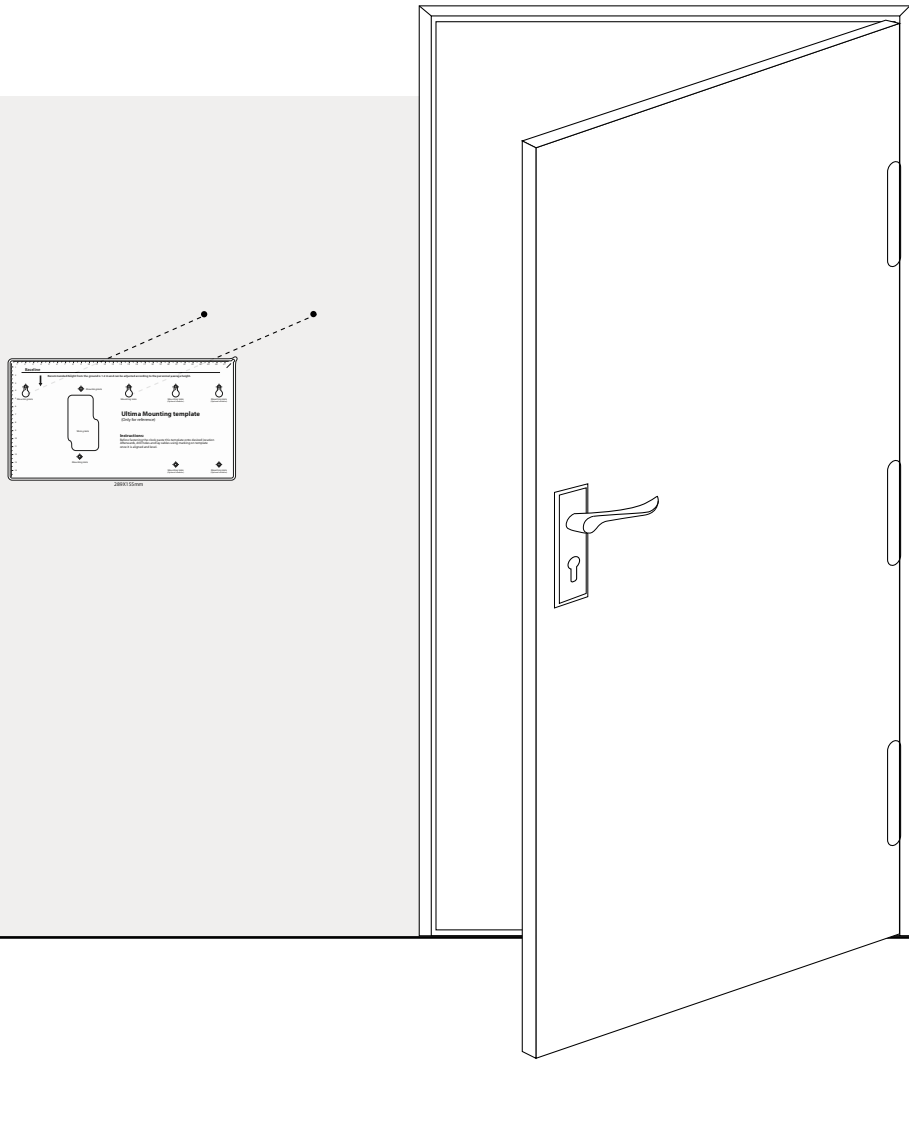
Note: The mounting template contains four fixing holes for the optional modules (finger-print & RFID module or RFID module, barcode/magnetic module), if you don't choose these modules, then you can just mark the left four fixing holes for the base.



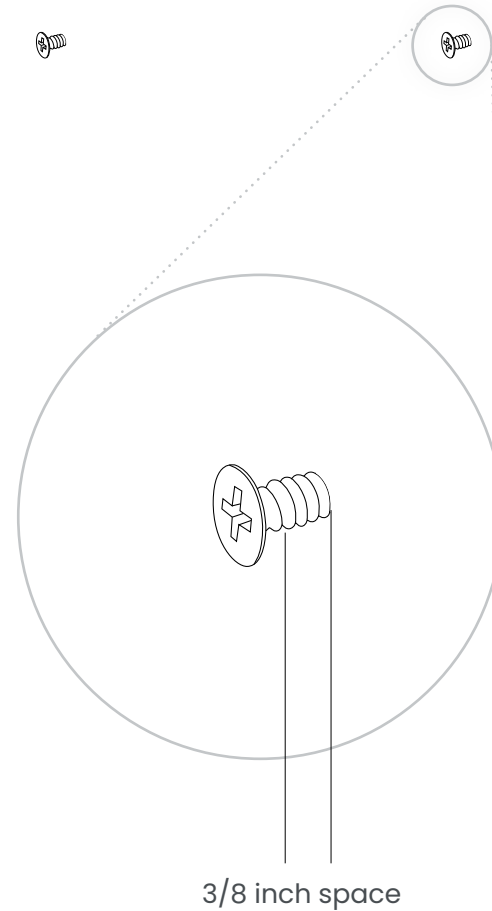
Drilling holes and fixing plastic anchors and upper screws



Drill 1/4 inch holes and insert plastic anchors flush with the wall surface.



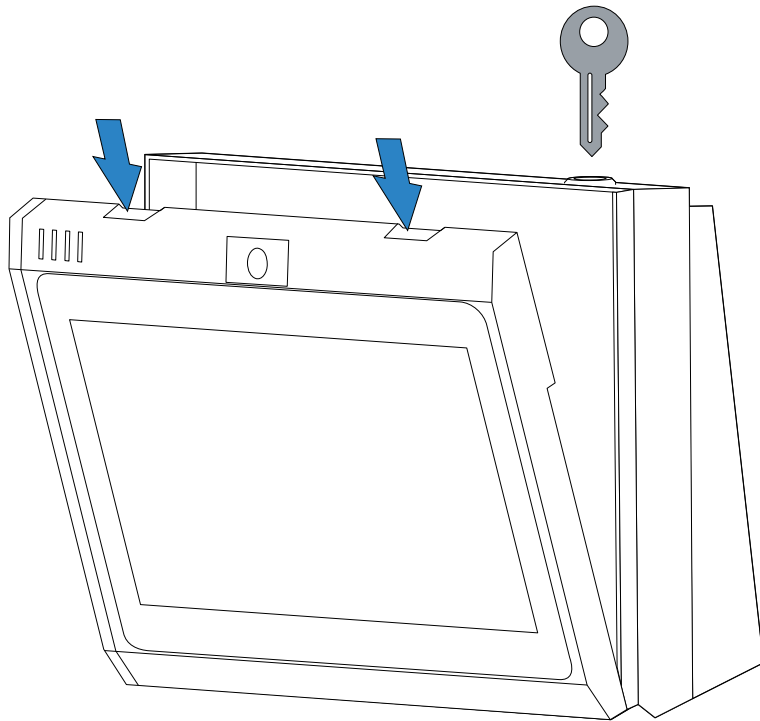
Fix the upper 2 screws (base only installation) on the wall.



Opening Ultima 7 front cover



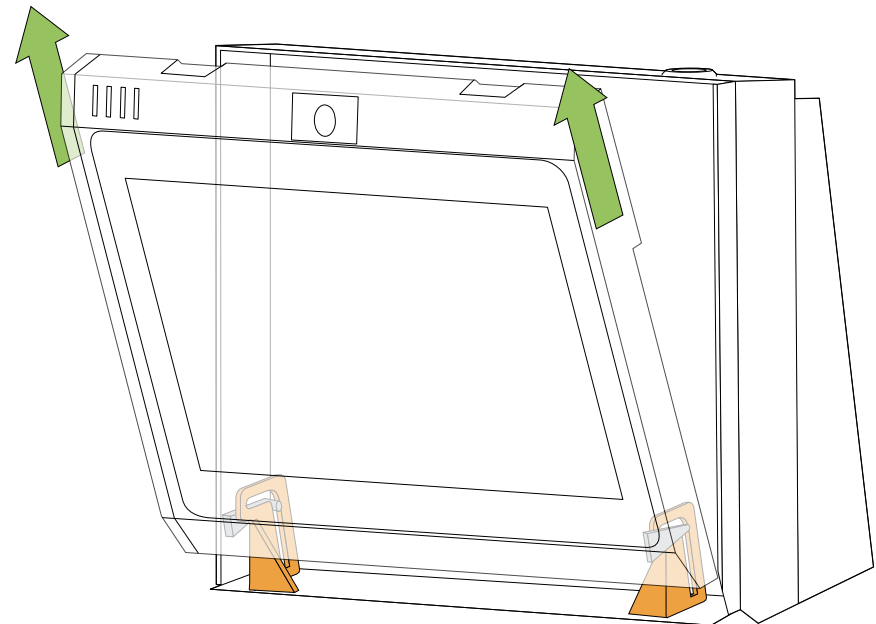
Step 1



Press down the tabs simultaneously and gently open it about 1.5 inch.

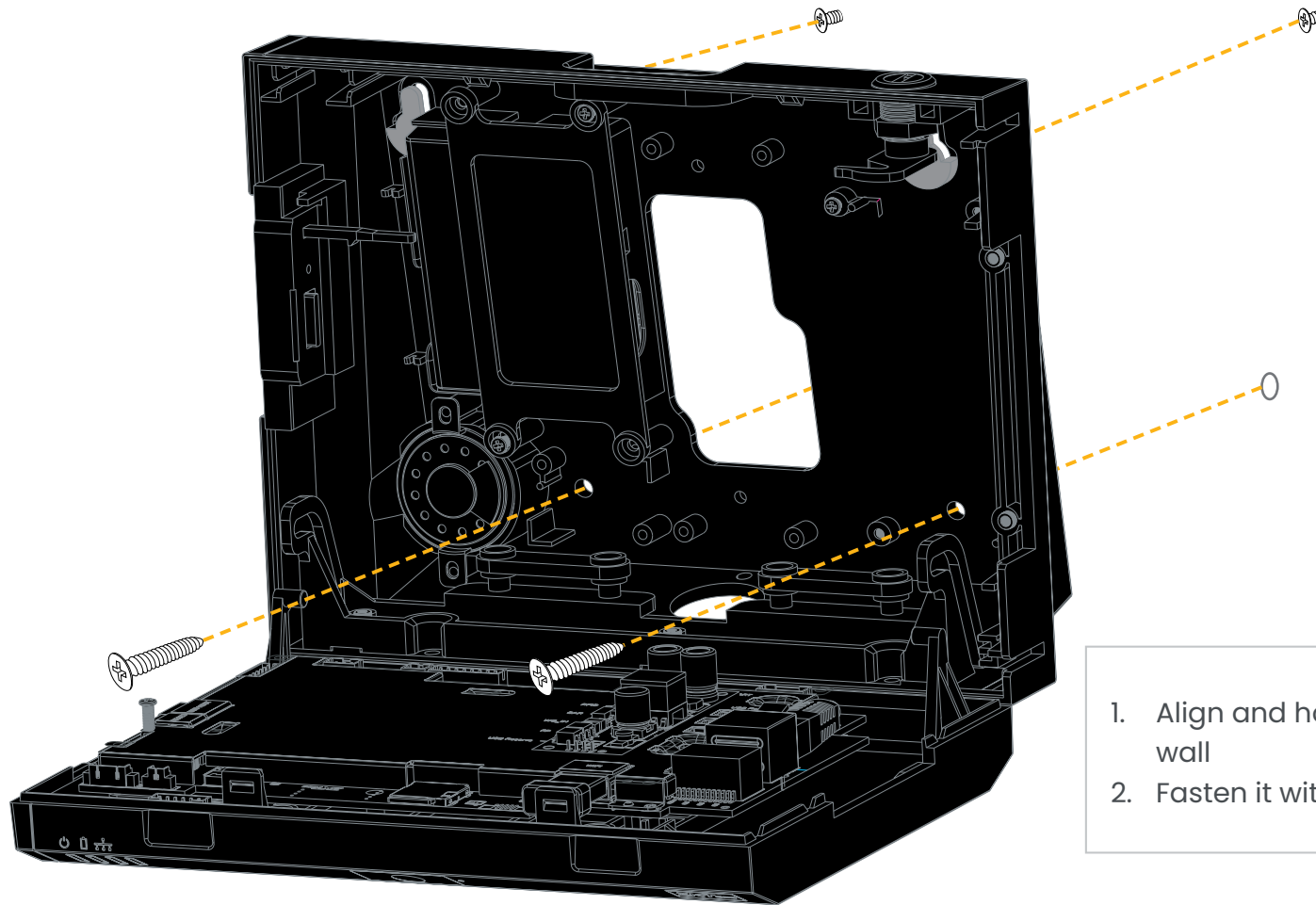
Caution Do not apply force

Step 2



Pull the door assembly UP about 1 inch to open it completely.

Caution Do not apply force



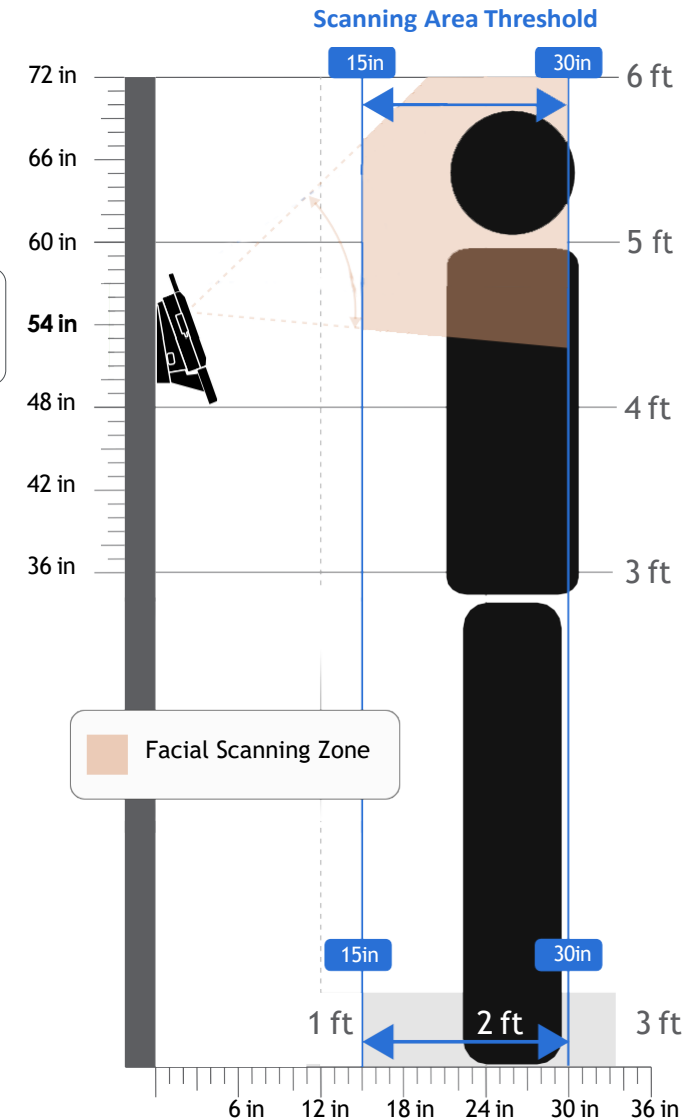
1. Align and hook up the Ultima 7 on to the wall
2. Fasten it with 2 bottom screws

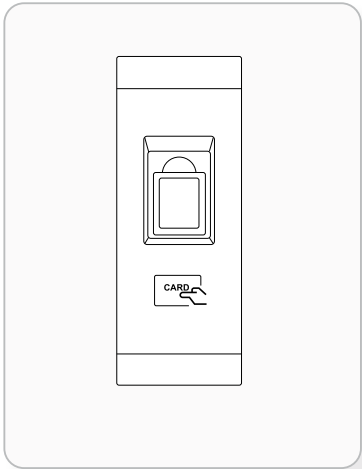


FOV (Field of view) for the combined facial sensors:

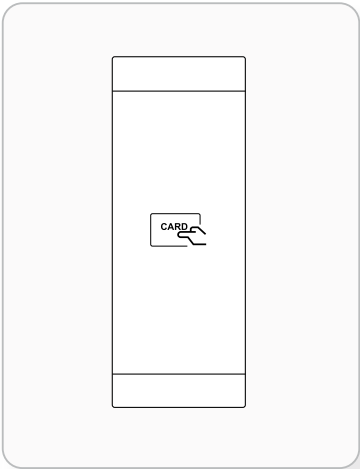
1. Per ADA regulations, ZKTeco WFM recommends installing the Ultima 7 clock between 54 inches from the top screw to 48 inches from the bottom above the ground
2. The “strike zone” in which the user should present their face for identification should be between 20 to 30 inches from the clock
3. The field of view for combined facial sensor fits anyone with a height of 4’08” to 6’00” while standing up straight between 20 to 30 inches in the “strike zone” from the clock
4. Anyone with a height of 6’00” to 6’06” should bend at their knees and place their face within the strike zone which is between 20 to 30 inches to fit inside the camera’s view angle for facial identification

Align the template top screw holes at 54 inch from the floor surface.

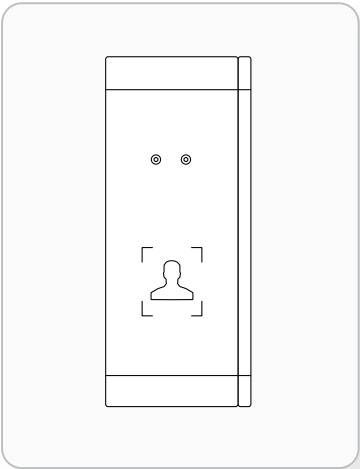




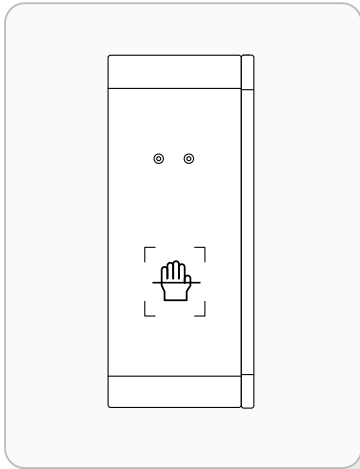
Fingerprint reader



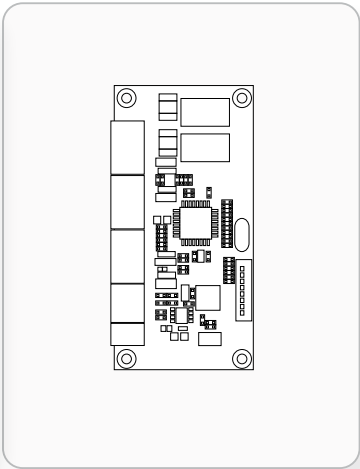
RFID reader



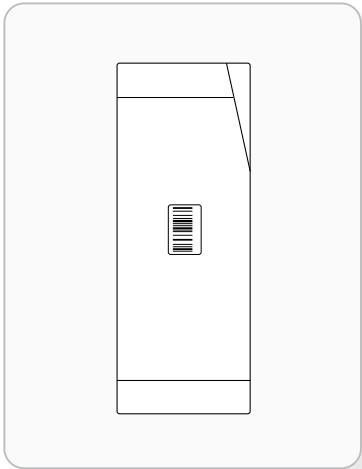
Facial recognition



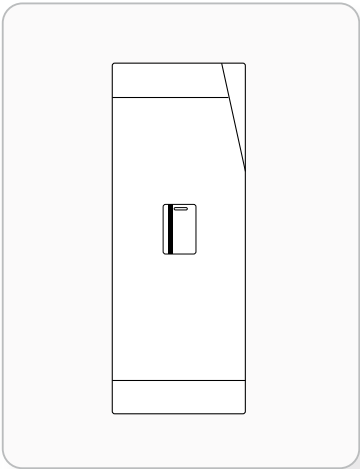
Palm recognition



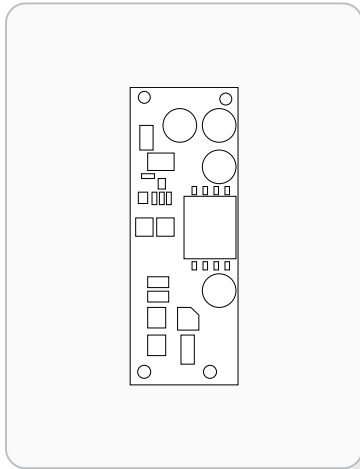
Relay



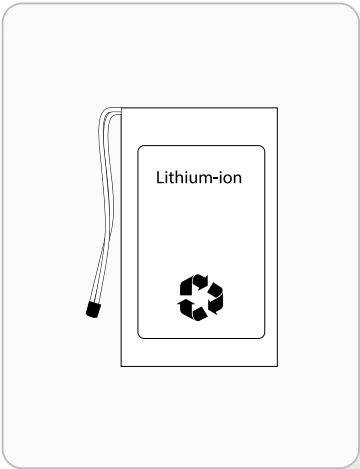
Barcode reader



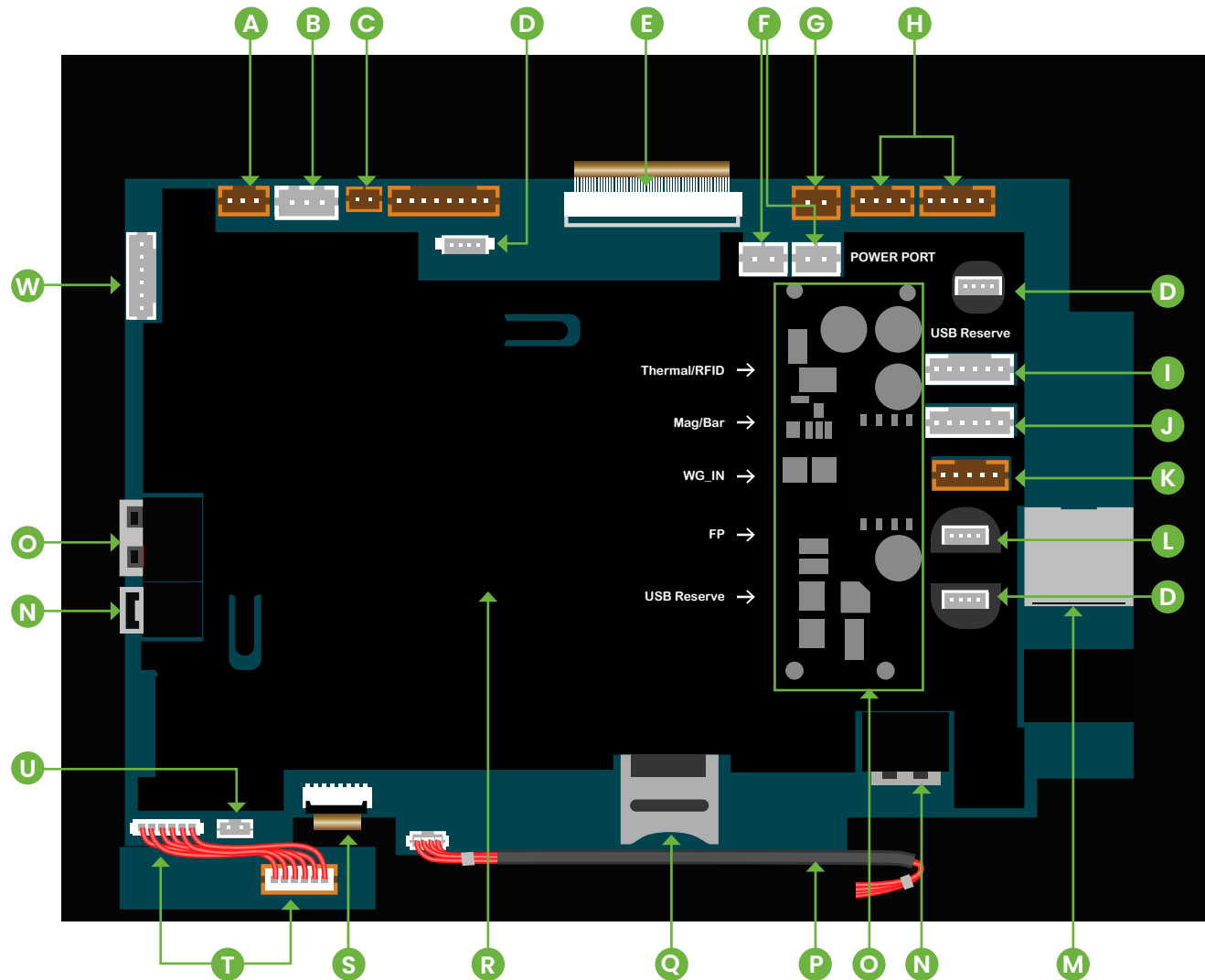
Magnetic reader



PoE+



Battery pack

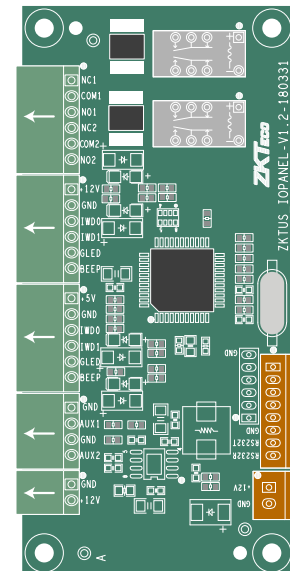
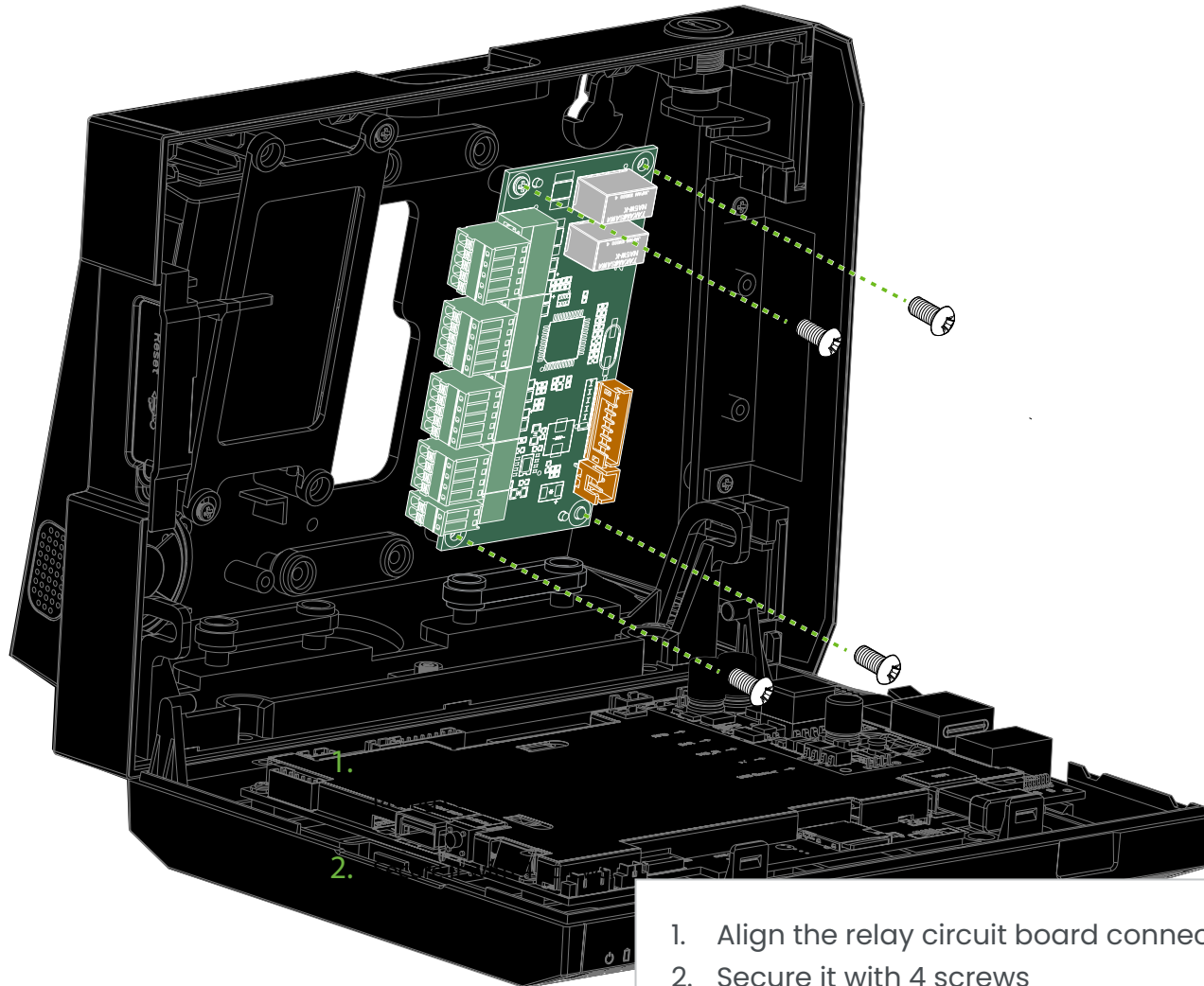


- A.** RS485
- B.** Battery
- C.** Speaker
- D.** USB Reserve
- E.** Display
- F.** 12 V 2A Power OUT
- G.** 12 V 3A Power IN
- H.** Relay Module
- I.** RFID Module
- J.** Barcode/Magnetic Reader
- K.** Wiegand IN
- L.** Fingerprint Module
- M.** Ethernet Port
- N.** USB Port
- O.** PoE+ Module
- P.** Camera
- Q.** SD Card
- R.** Heat Sink
- S.** Touch Screen Sensor
- T.** LED Status indicator
- U.** Microphone
- V.** Reset
- W.** GPS

Installing the relay module



Place the relay module in the back shell and fix it with 4 screws.



1. Align the relay circuit board connectors facing center
2. Secure it with 4 screws

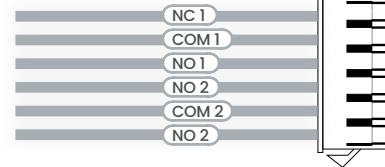
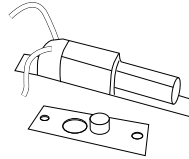
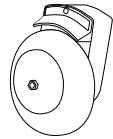
Relay module connectors



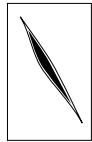
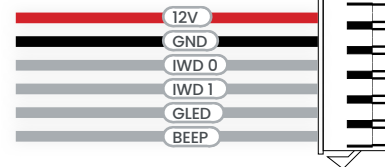
Warning Relay is rated for **12V only** and **should not exceed the recommended voltage**. If connecting a higher voltage device some extra components maybe necessary. Please consult ZKTeco WFM for further guidance before connecting to higher voltages.

Connecting wires

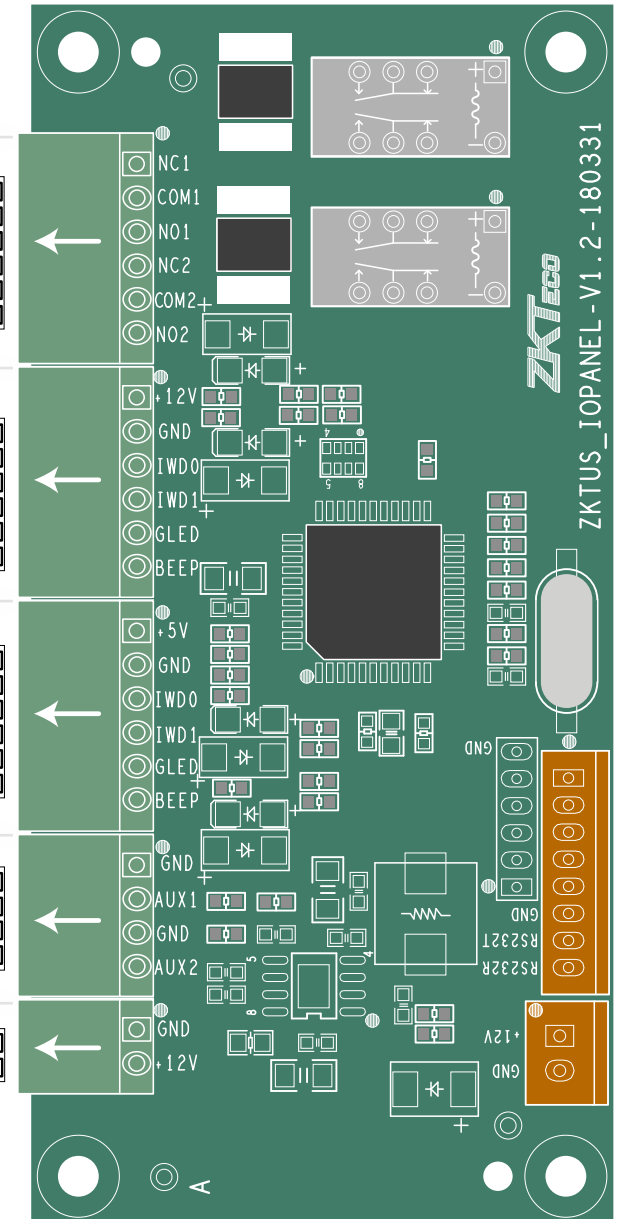
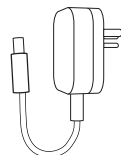
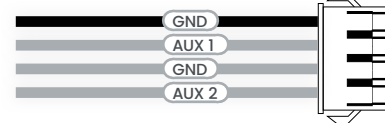
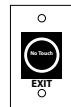
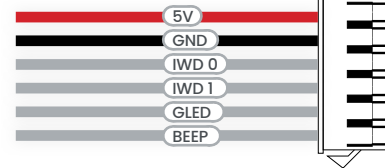
Please follow components manufacturer installation instructions.



12 Volt Components



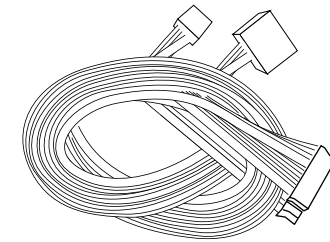
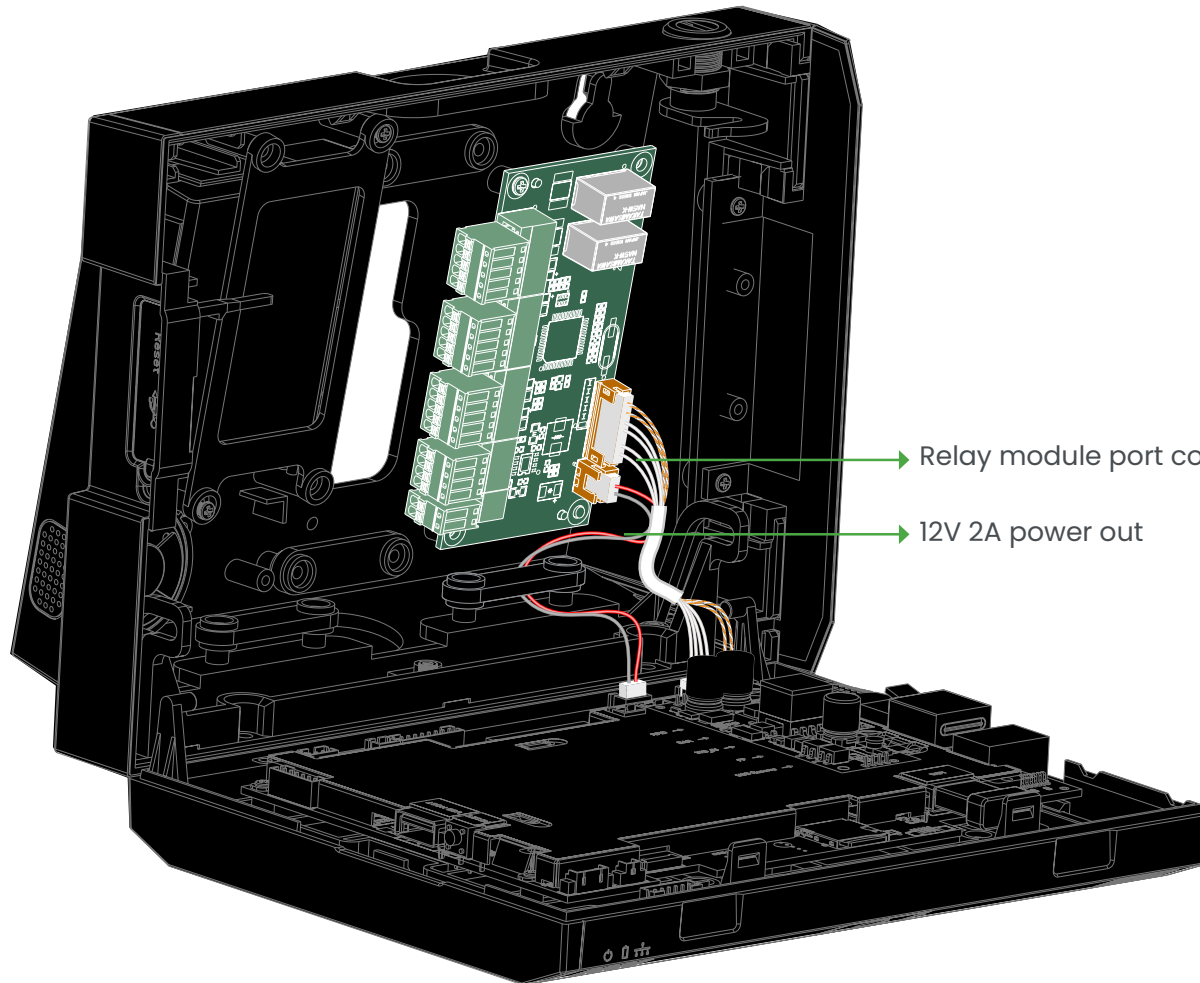
5 Volt Components



Connecting the relay module



Connect the cables between the relay module and the main board.

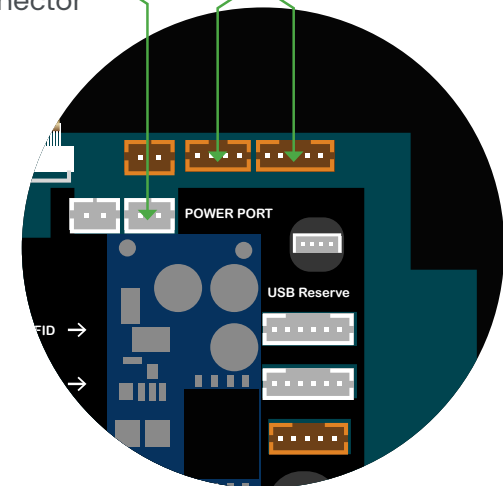


12V 2A power
connector

Relay module
port connector

Relay module port connector

12V 2A power out



Motherboard

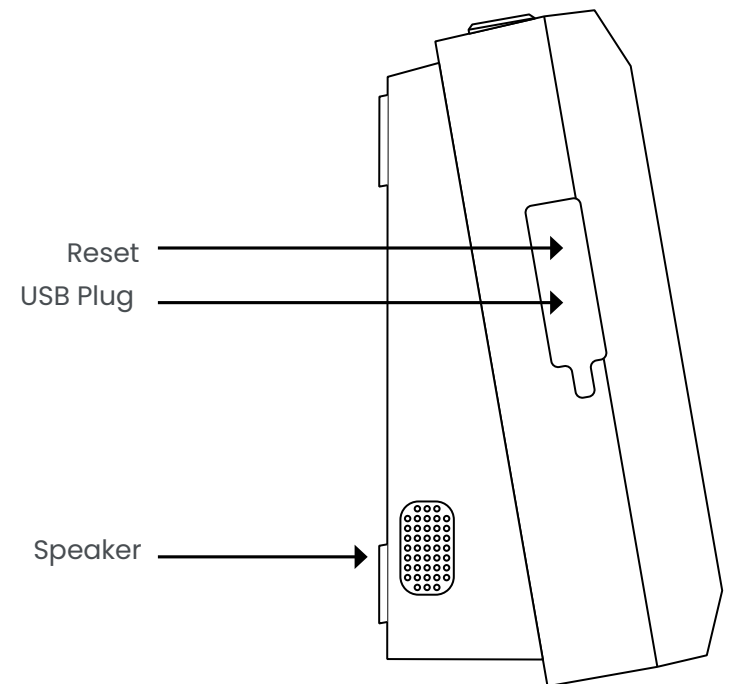
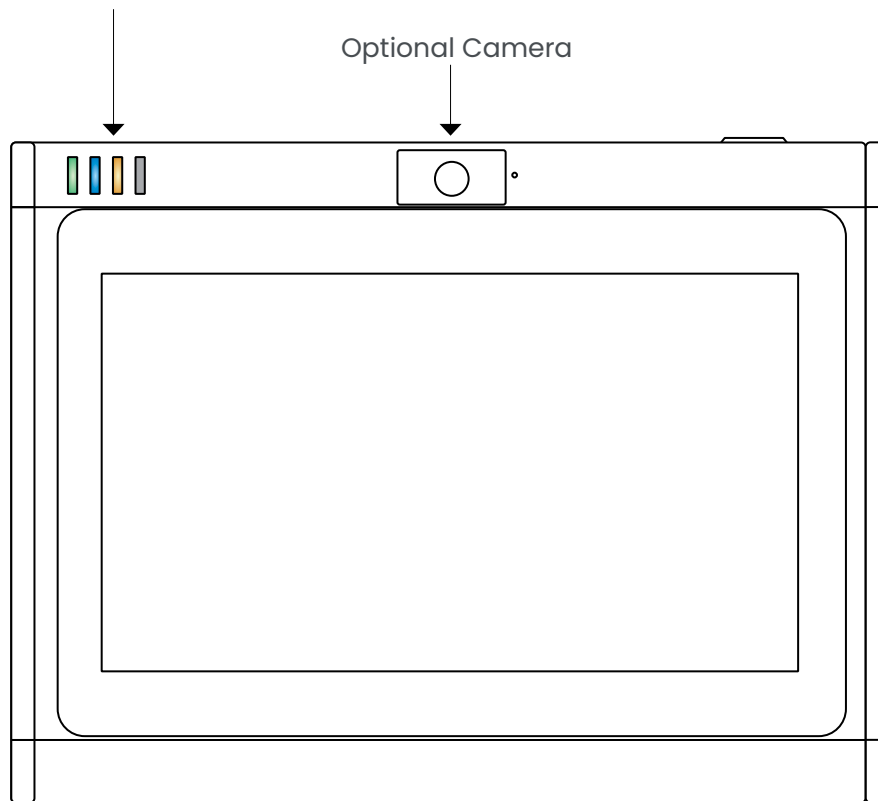


Warning Relay is rated for **12V only** and **should not exceed the recommended voltage**. If connecting a higher voltage device some extra components maybe necessary. Please consult ZKTeco WFM for further guidance before connecting to higher voltages.

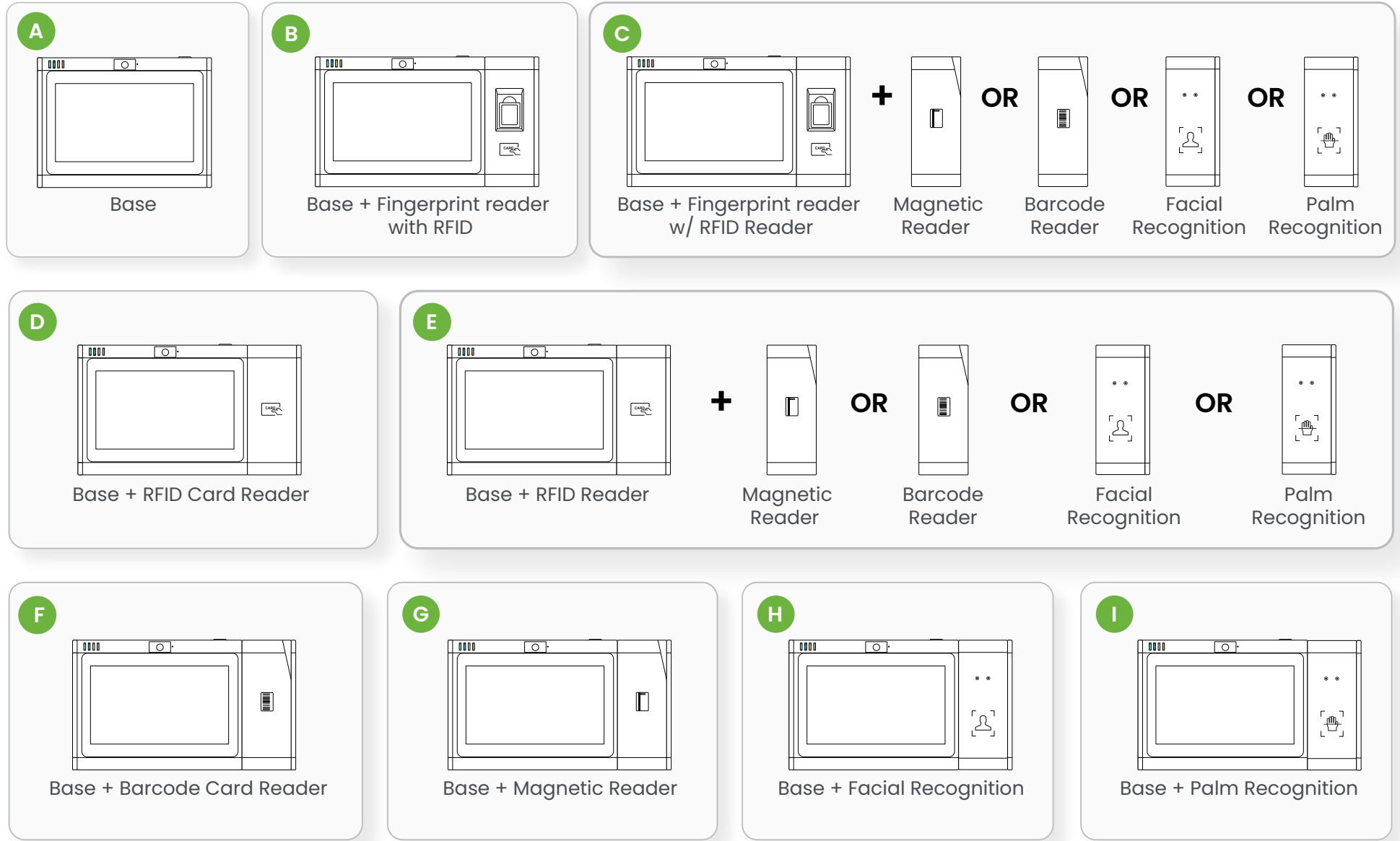
External plug and indicators

LED Indicators

- AC Power
- Battery Power
- Network
- Authentication Verification



Ultima 7 configuration options



Installing a single module

B

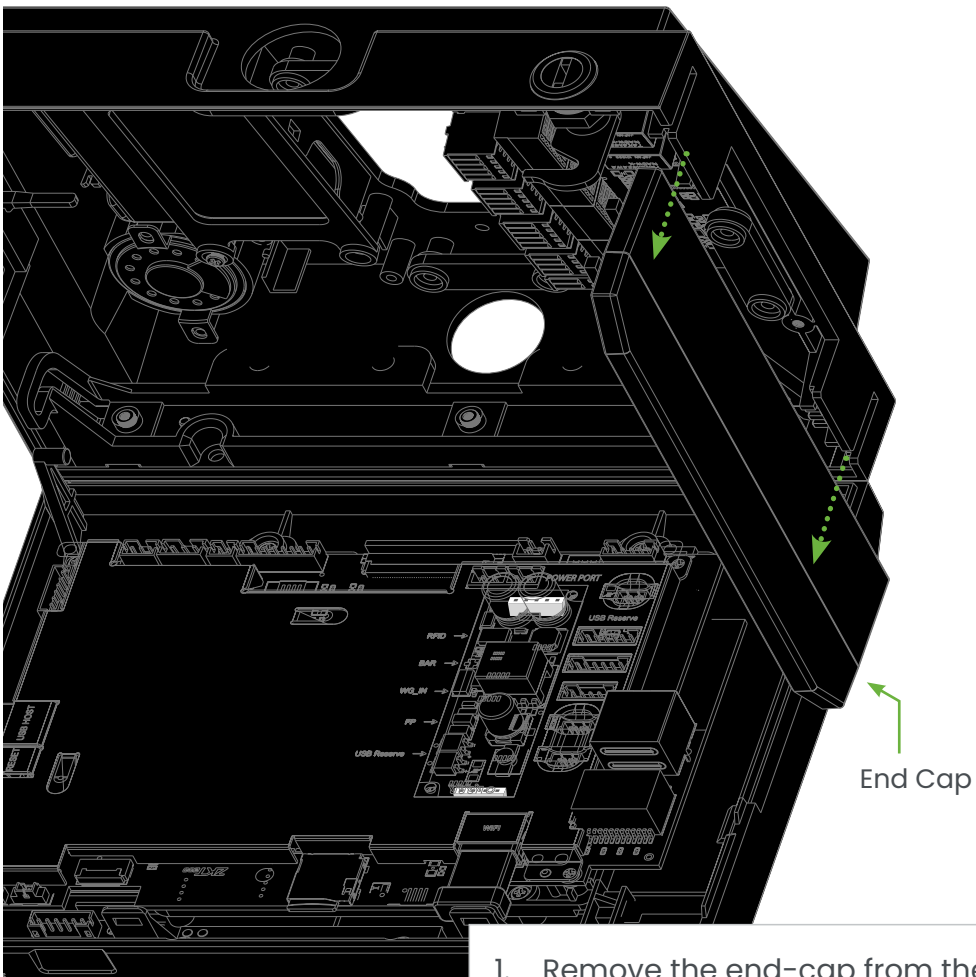
D

F

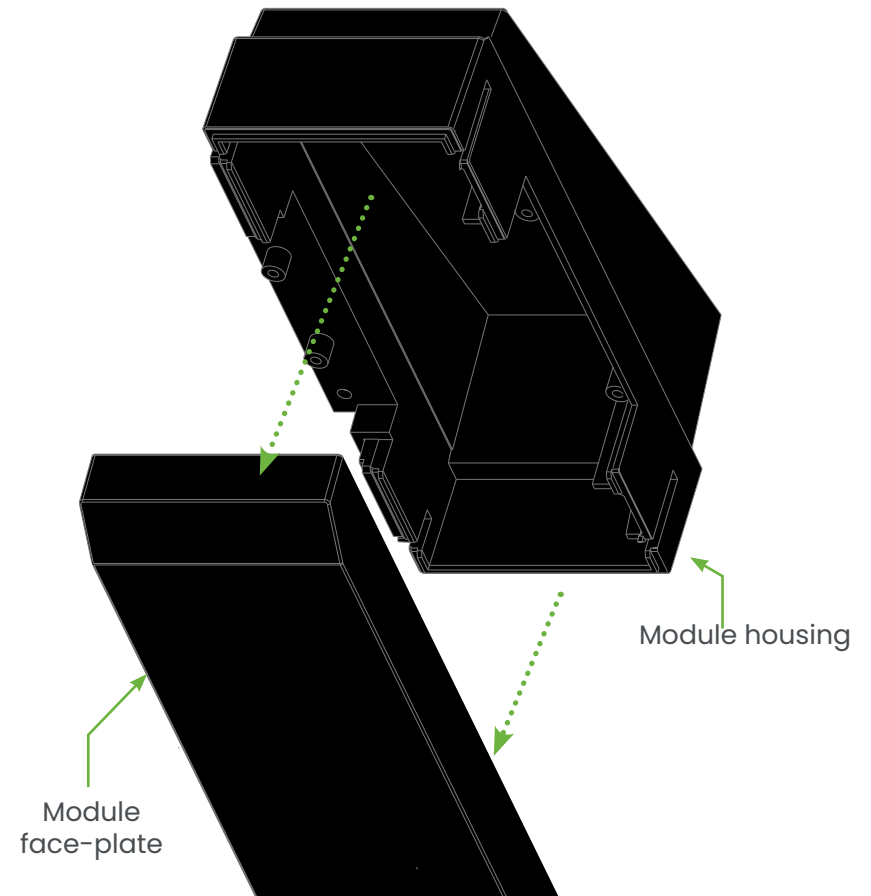
G

H

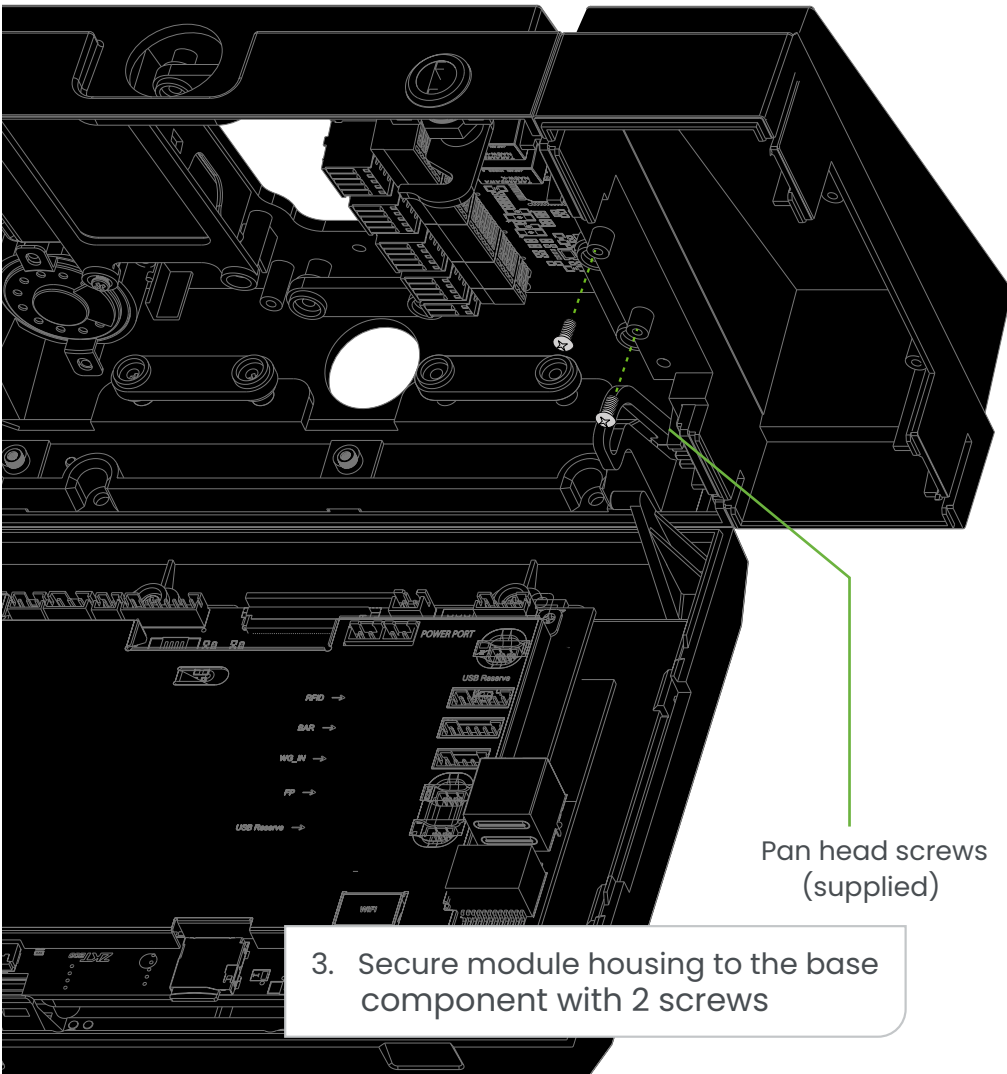
I



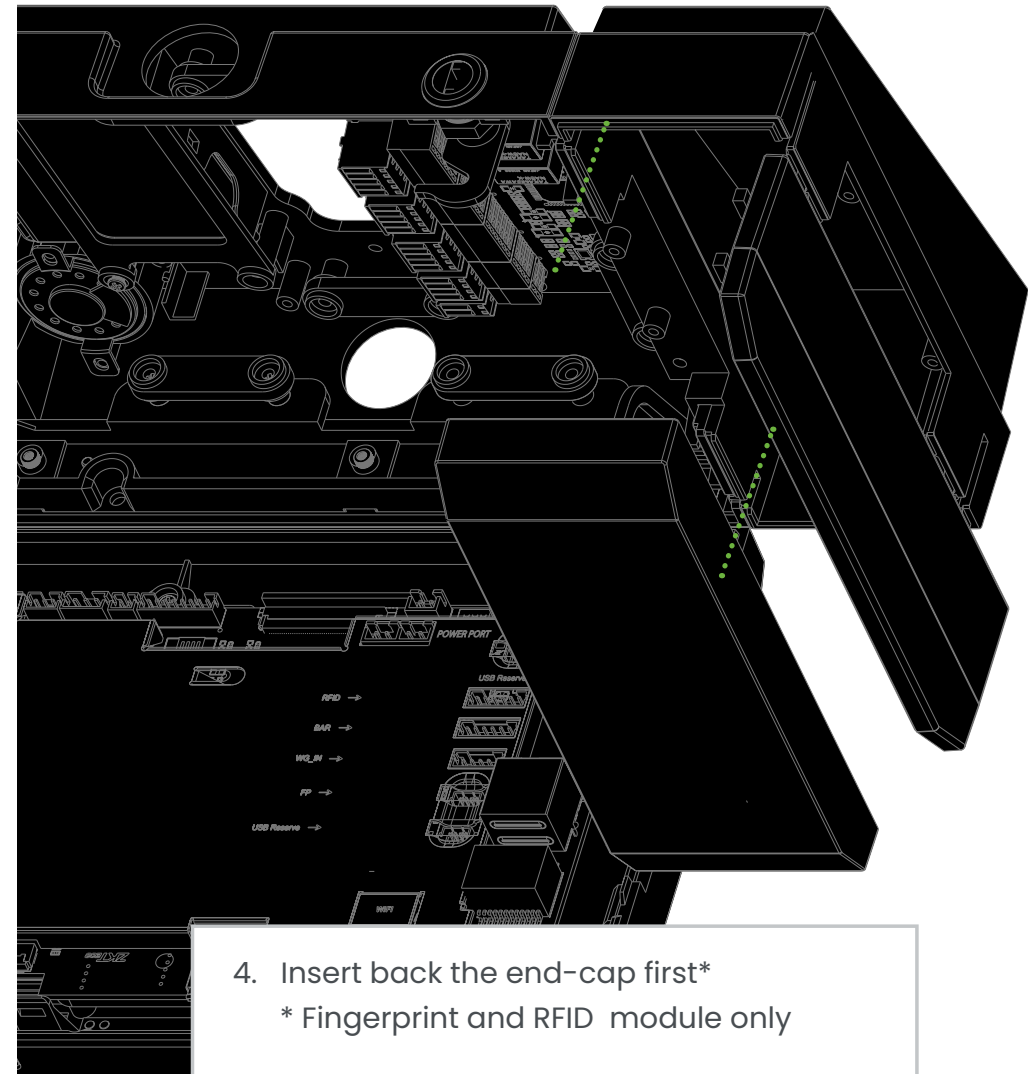
1. Remove the end-cap from the base by sliding it out forward of the base unit



2. Remove faceplate from the housing by pulling it forward (apply gentle force)



- Secure module housing to the base component with 2 screws



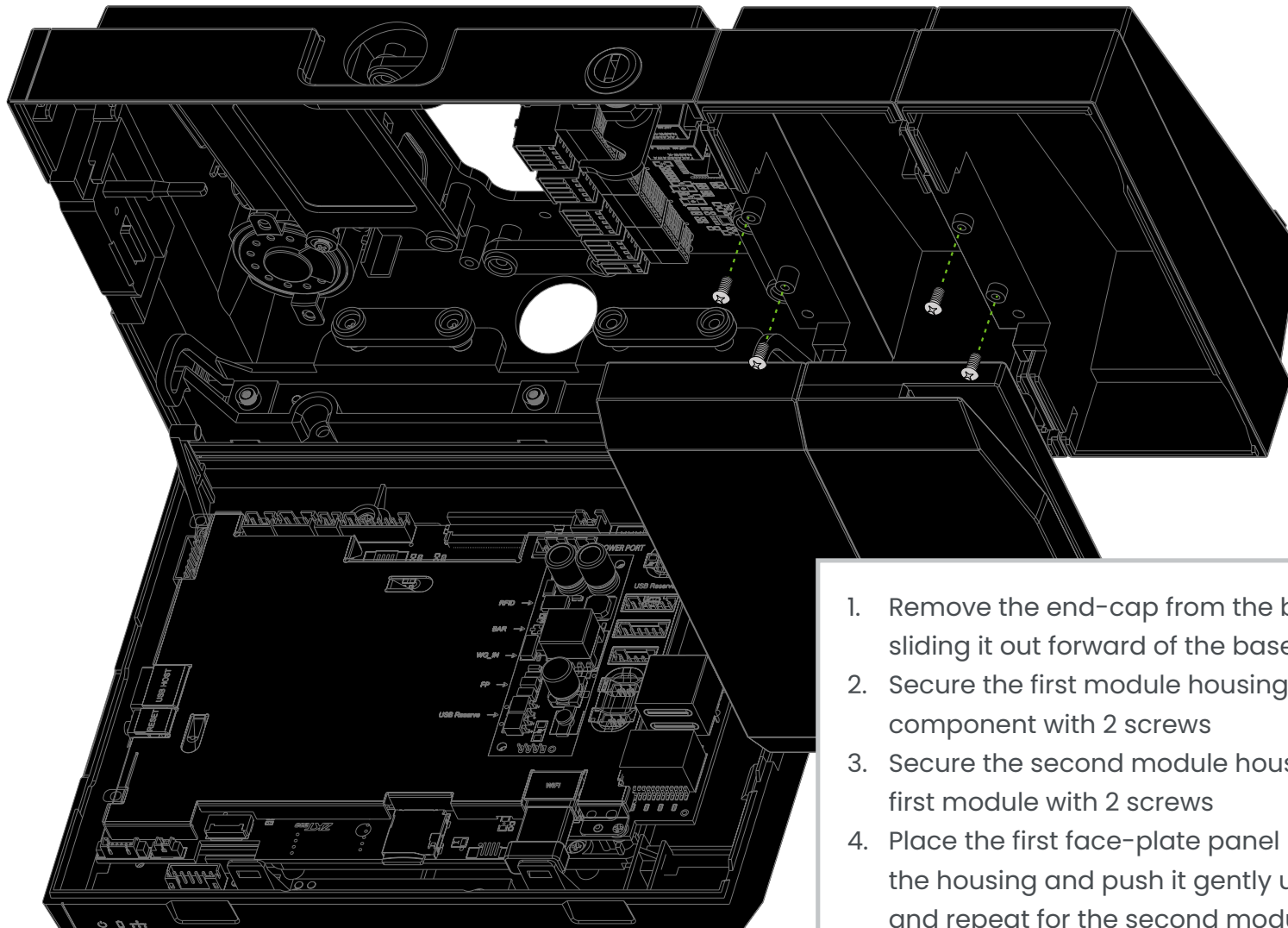
- Insert back the end-cap first*
* Fingerprint and RFID module only
- Place the face-plate panel back on to the housing and push it gently until it snaps

Installing dual modules

C

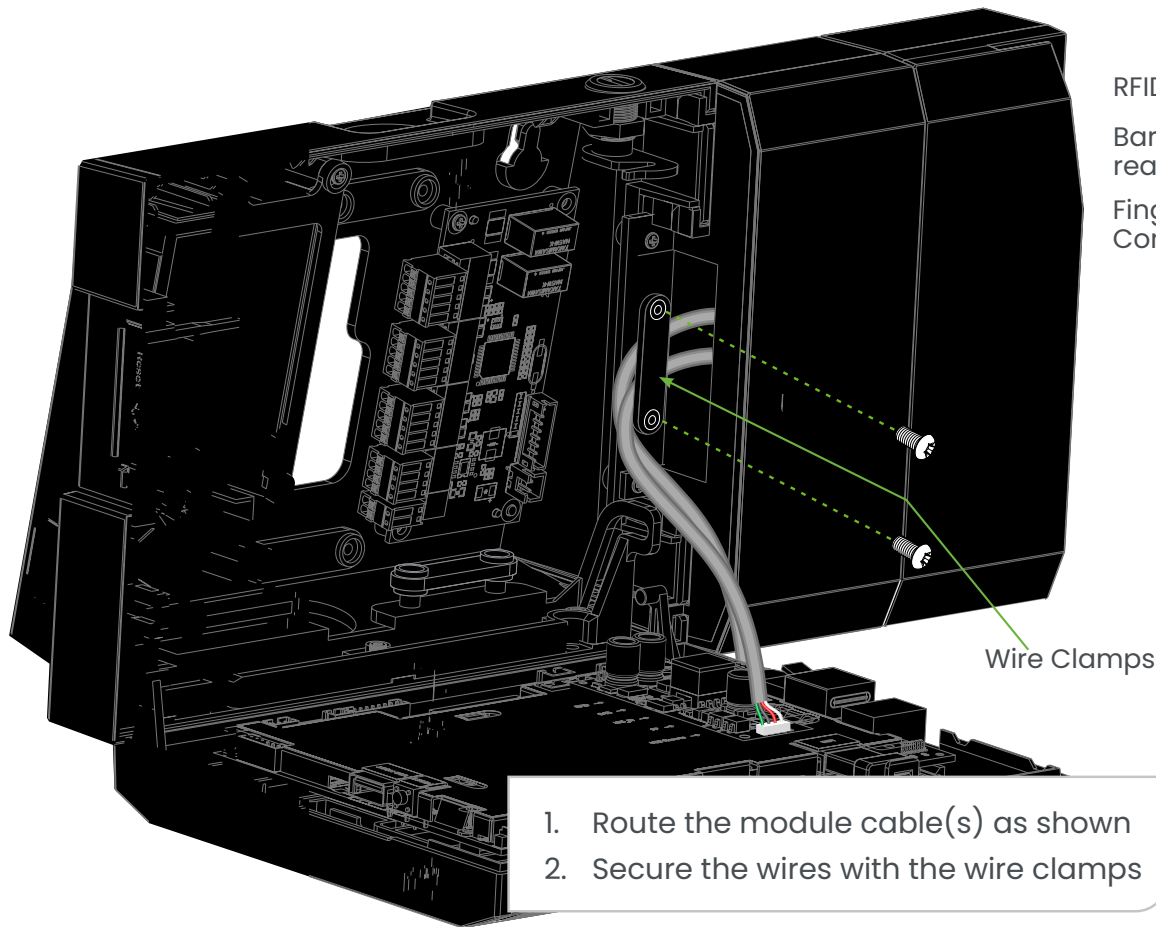
E

Fingerprint or RFID module must be installed first.



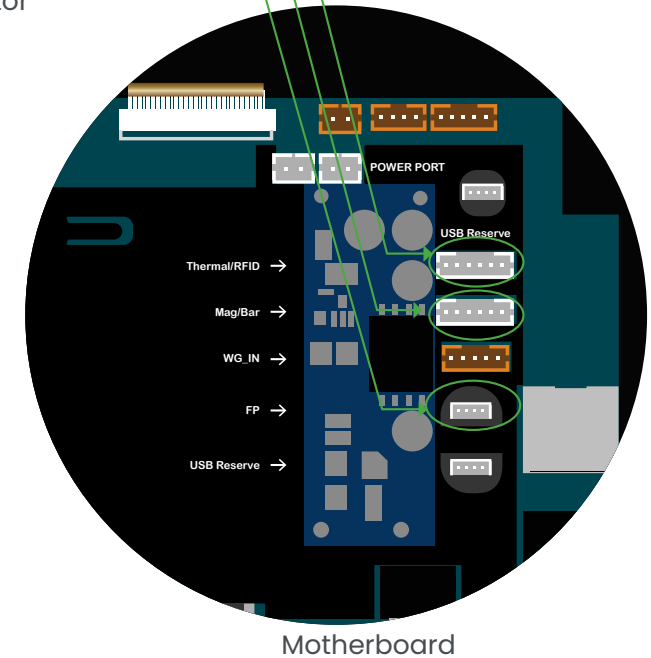
1. Remove the end-cap from the base by sliding it out forward of the base unit
2. Secure the first module housing to the base component with 2 screws
3. Secure the second module housing to the first module with 2 screws
4. Place the first face-plate panel back on to the housing and push it gently until it snaps and repeat for the second module

Connecting the module wires to the motherboard



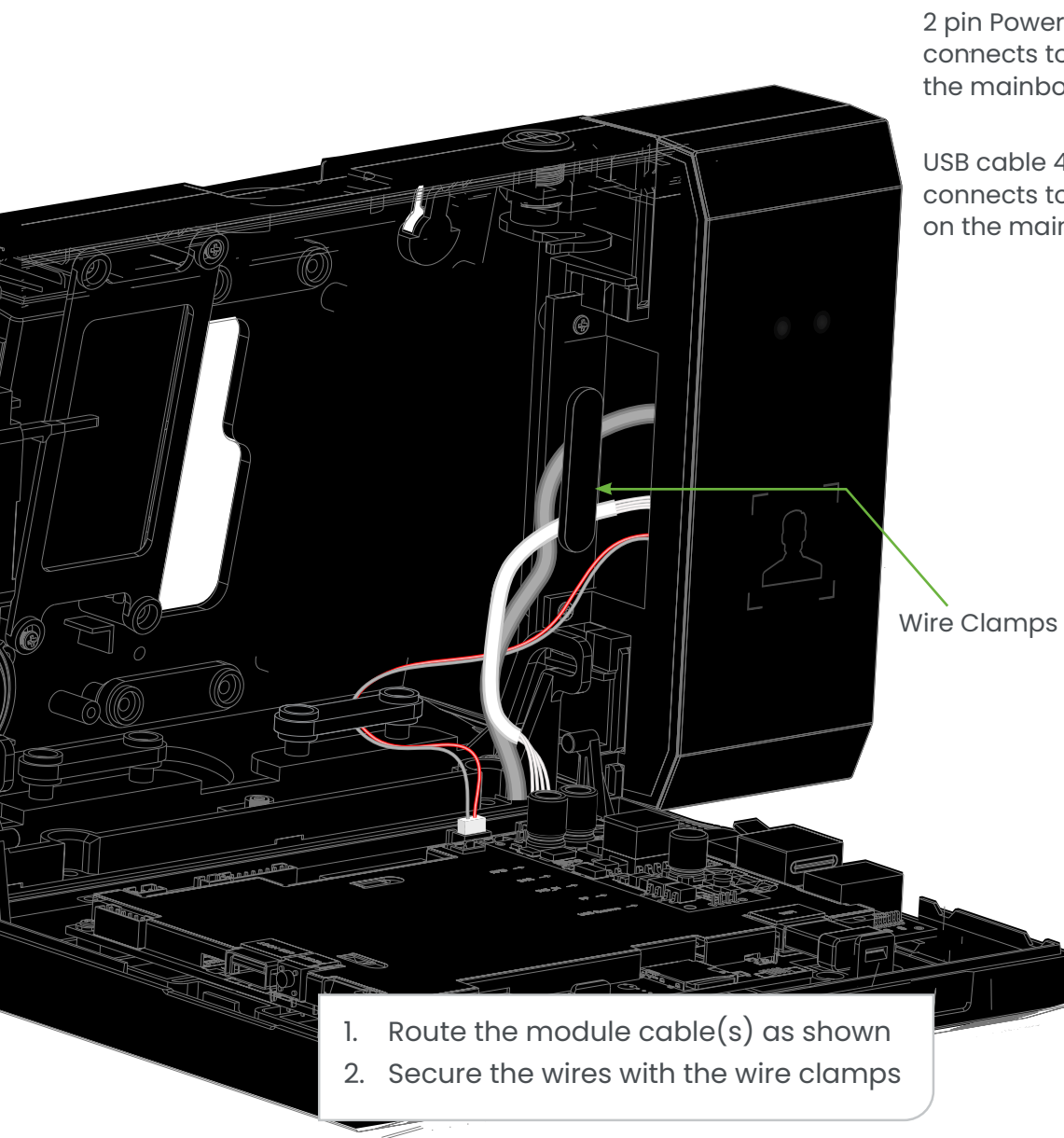
RFID Connector
Barcode & Magnetic card
reader Connector
Fingerprint
Connector

Wire Clamps



Motherboard

Connecting the Facial module wires to the motherboard

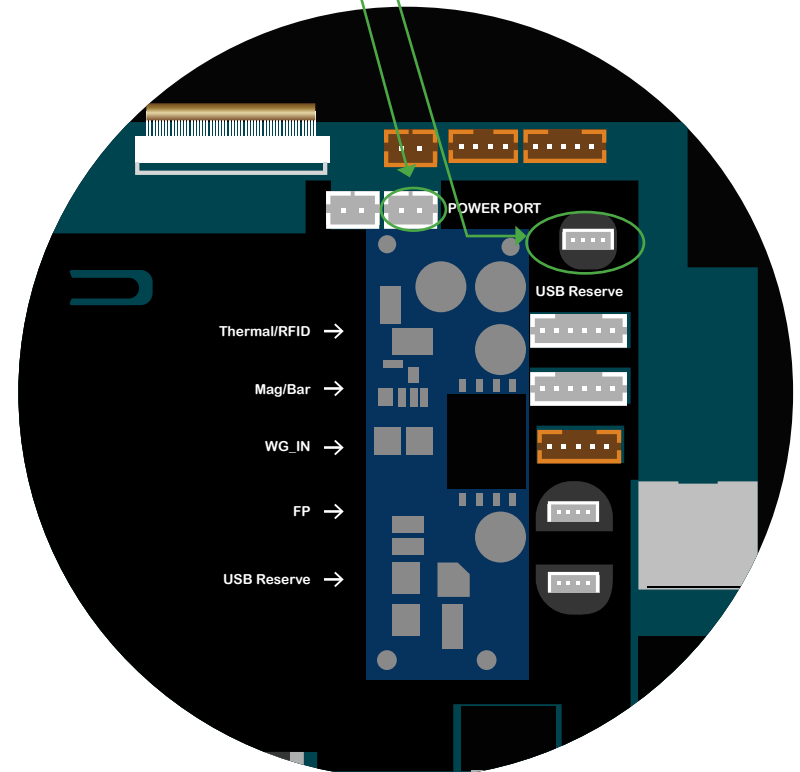


2 pin Power supply 12V which connects to white port J3 on the mainboard

USB cable 4 pin for Face module connects to port J24(USB Reserve) on the mainboard

Wire Clamps

1. Route the module cable(s) as shown
2. Secure the wires with the wire clamps

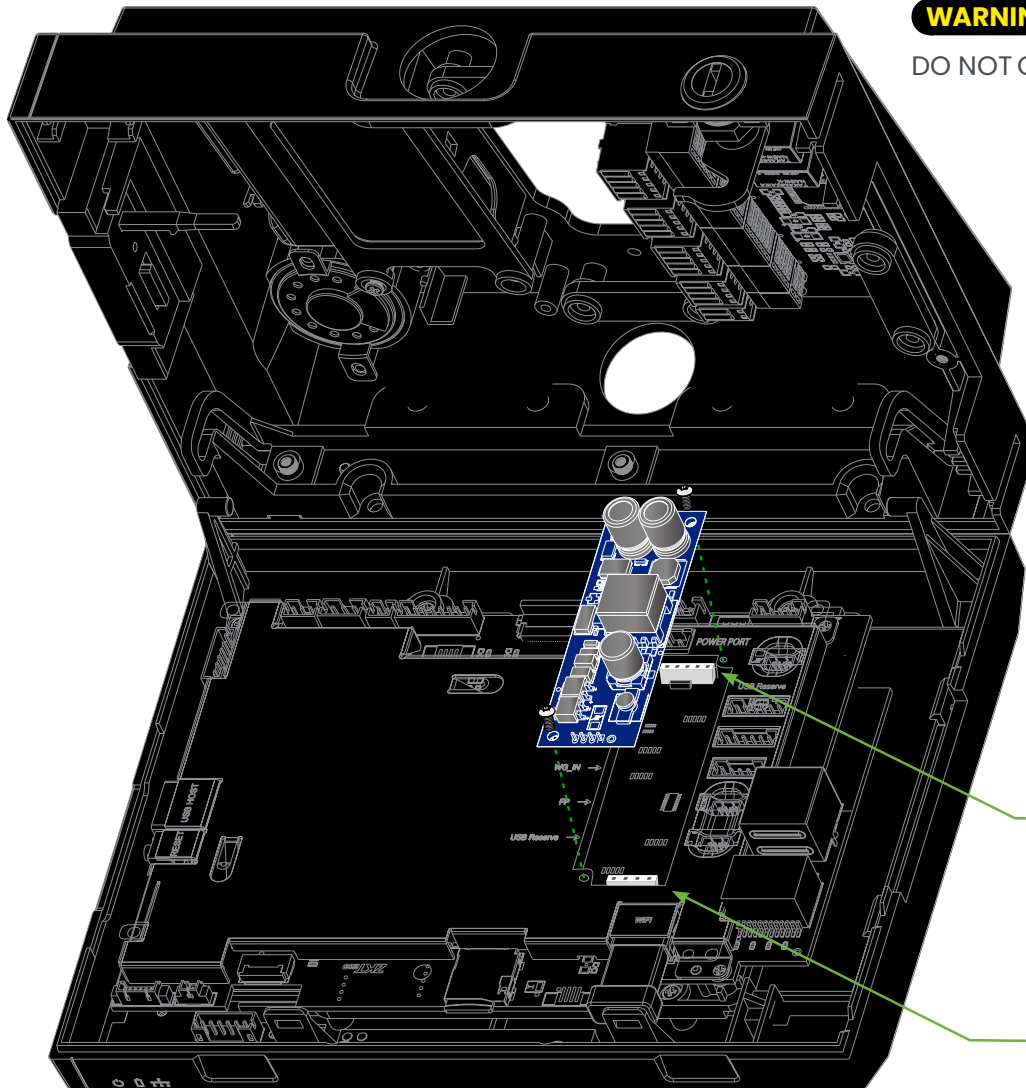


Motherboard

Installing internal PoE+ module (Power Over Ethernet+)



WARNING Power up your device AC Adapter **OR** Power-over-Ethernet+ (PoE+)
DO NOT Connect simultaneously (It may damage the device)



1. Align 5 pin and 4 pin connectors to corresponding plug connectors
2. Gently push-in to connectors and secure it with 2 screws

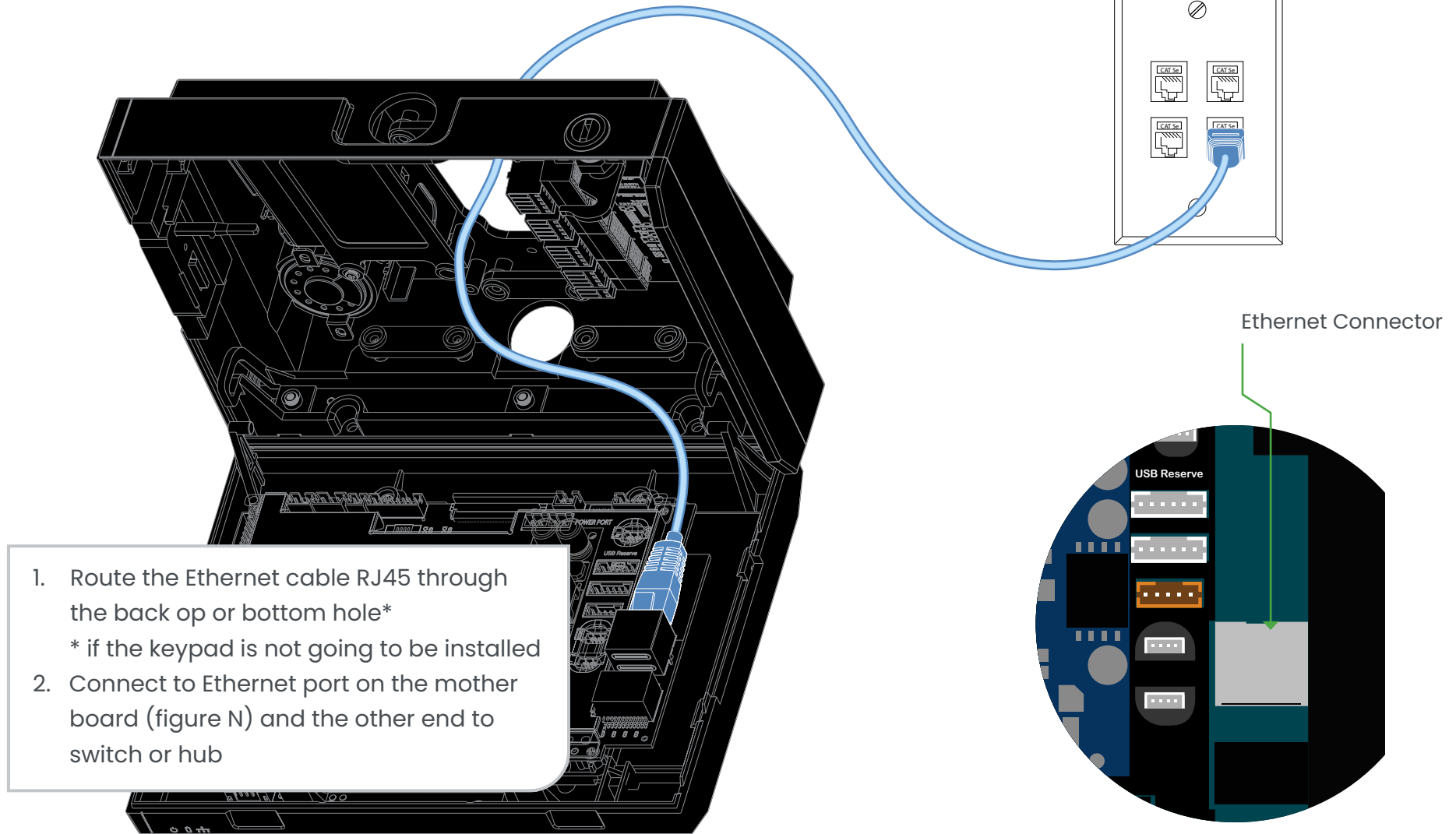
5 pin Connector

4 pin Connector

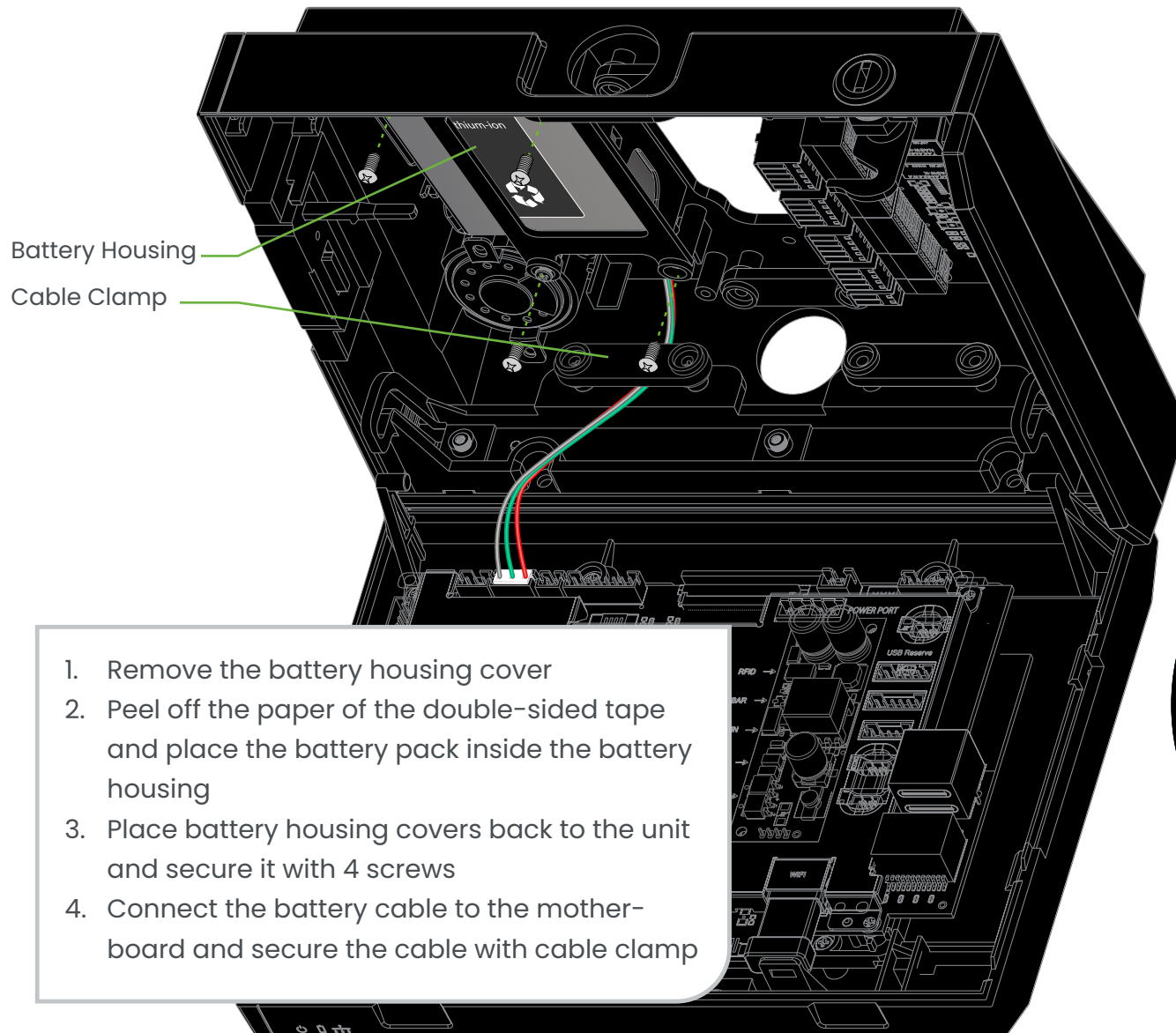
Connecting Ethernet Cable



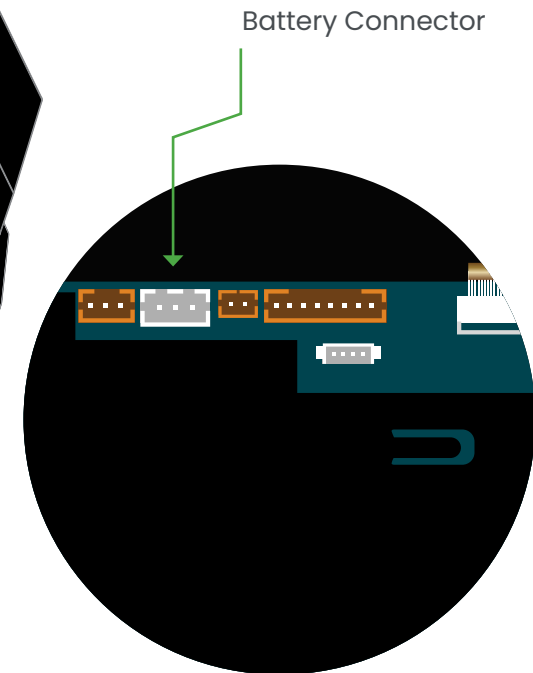
WARNING Power up your device AC Adapter **OR** Power-over-Ethernet+ (PoE+)
DO NOT Connect simultaneously (It may damage the device)



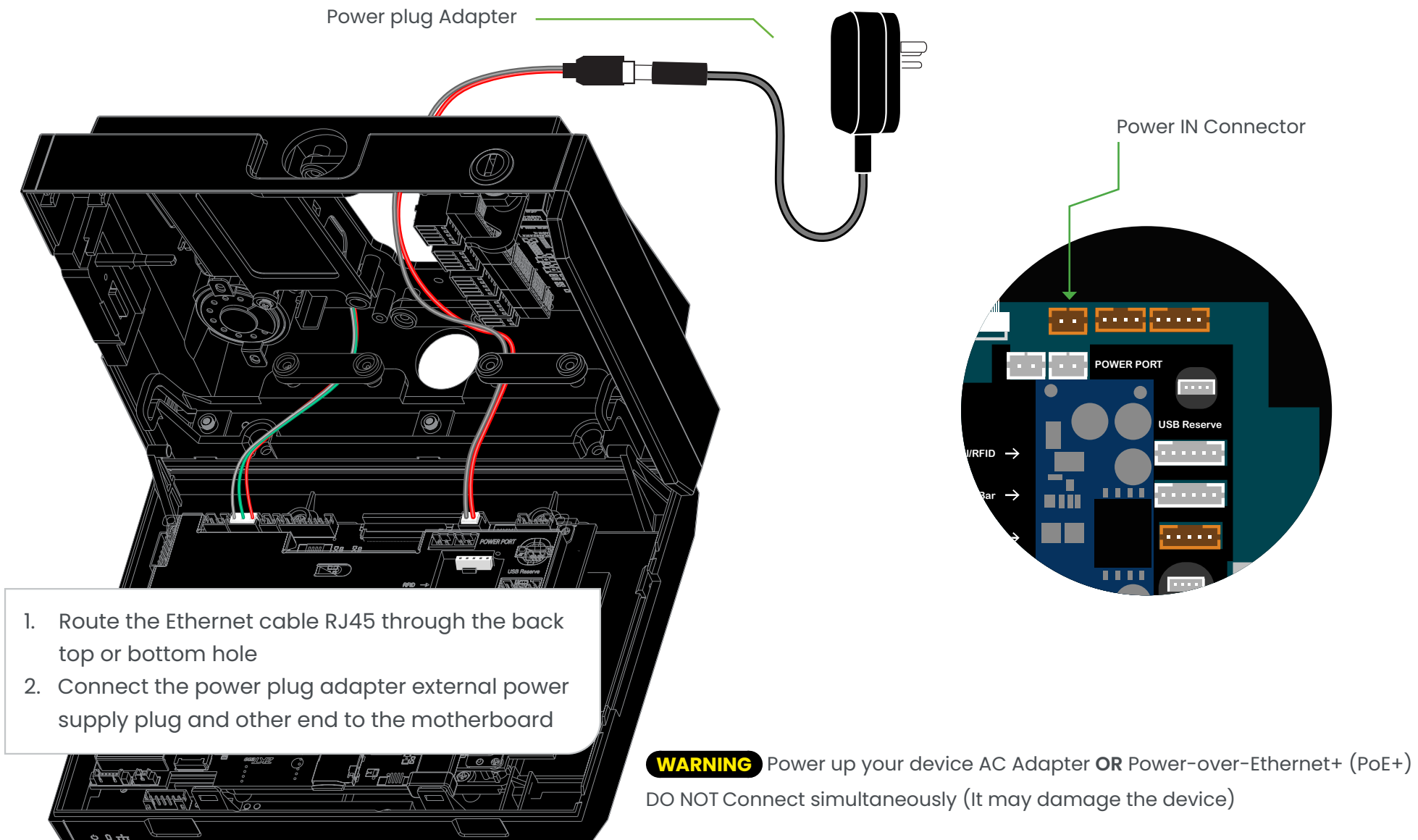
Installing Internal Battery Pack



1. Remove the battery housing cover
2. Peel off the paper of the double-sided tape and place the battery pack inside the battery housing
3. Place battery housing covers back to the unit and secure it with 4 screws
4. Connect the battery cable to the motherboard and secure the cable with cable clamp



Connecting AC power supply

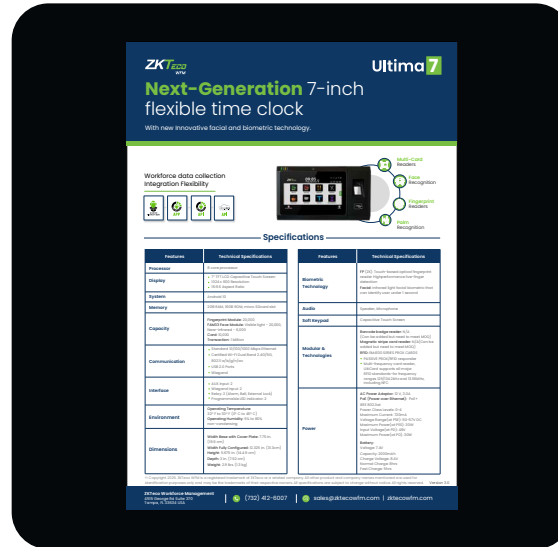




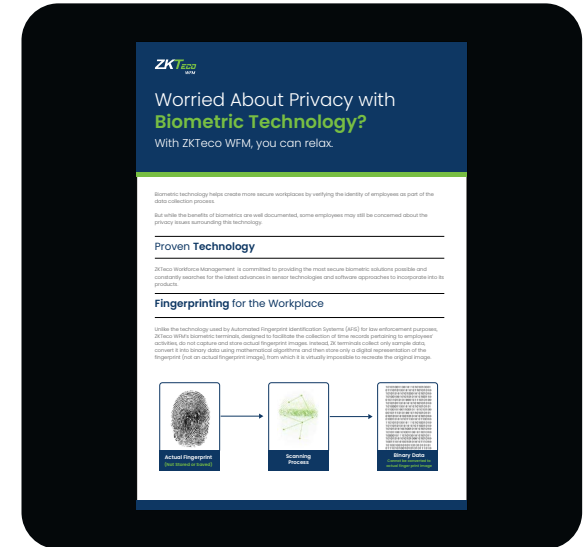
Issue	Probable Causes	Corrective Action
There is no power in the Ultima 7	If the Ultima 7 is powered using standard AC power cable, the power supply is not properly plugged into the main board, the AC outlet, or both	Check the connections to the main board and the AC outlet Note: Do not plug the power cord into the AC outlet until you ensure that the power supply is properly connected to the main board
	If the Ultima 7 is powered using Ethernet cable, the network may be down, or connections to and from the PoE+ module may not be secure	• Ensure that the network is running • Ensure that the network cable is connected to the main board
The power LED is turned on, but nothing appears on the screen	If the Ultima 7 is powered using a standard AC power cable, the AC line is not live or is supplying improper voltage	Measure voltage at the AC outlet and, if necessary, locate another power source
	If the Ultima 7 is powered using Ethernet cable, there is an issue with the network	• Check the Ethernet connection at the Ultima 7, and all other network connections • Ensure that the network is supplied with an 802.11af compliant power source
Communication fails	Communication settings are incorrect either at the Ultima 7 or the host PC	Check and, if necessary, correct the device IP Address, Netmask, Gateway and DNS.
	Communications cable is connected incorrectly	Ensure that all communications cables are routed correctly and connected properly



Fingerprint Enrollment
Matching Information



Datasheet



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